

Moon Formation by Triple Phase Transition in the Differentiating Proto-Earth

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Abstract. The Moon is there. This is a fact, not a hypothesis to be confirmed: our task is to trace back its causes. Just as our own existence attests that the improbable chain of conditions that made it possible really did occur, the existence of the Moon — with its mass, its isotopic identity, and the angular momentum of the Earth–Moon system that we observe — imposes a precise set of conditions on the state of the Hadean proto-Earth. This theory does not seek to tune free parameters to reproduce a desired outcome: it proceeds by *reconstruction*. The values we derive from it (radial density gradient, obliquity, angular momentum input) are not chosen; they are constrained by an outcome that is itself certain. That a window of conditions is narrow is therefore not a weakness, but a retrodictive prediction about the early Earth.

Several well-established features of the Earth–Moon system — the near-identical Earth–Moon isotopic composition (Wiechert et al., 2001; Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025), the crustal dichotomy (Wieczorek et al., 2013), and the ≈ 350 Ma delay of the terrestrial dynamo (Tarduno et al., 2025) — lack a fully satisfactory explanation within current lunar formation models. We propose an endogenous alternative, grounded in a necessary thermodynamic observation: any Earth-mass planet emerges from accretion in a state of near-total silicate mantle melting, the accretion energy exceeding the fusion energy by a factor ≈ 121 (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The proto-Earth is thus a rapidly rotating magmatic body ($T_{\text{rot}} \approx 3.5$ h), lacking a stabilizing satellite, whose axis undergoes large-amplitude free nutation within the interval $[40^\circ, 70^\circ]$ measured in its own co-rotating frame — an internal geometric oscillation of the fluid body (Laskar et al., 1993a; Touma & Wisdom, 1993; Li & Batygin, 2014).

A single engine — the progressive segregation of metallic iron-nickel toward the forming core — drives three coupled transitions. (i) A rheological transition: the magma acquires a Bingham-Herschel yield stress (Bingham, 1922; Herschel & Bulkley, 1926) and organizes a Coherent Magmatic Torus (CMT) within the intertropical band $|\phi| < 30^\circ$. (ii) A mechanical transition: the CMT undergoes $N = 2\text{--}3$ episodes of cohesive hypersonic ejection. Ejection proceeds from a **deterministic dynamical instability** — a parcel of the torus leaves it once the Solberg–Høiland stability coefficient $\lambda(U)$ becomes positive. The threshold is reached

cyclically: rise in angular momentum, ejection of a coherent magmatic tongue at nutation peaks, loss of momentum, recharge. Each ejection shifts the rotation axis toward a new obliquity, and therefore the active band itself: each successive CMT is geometrically distinct, underlying the observed crustal dichotomy. (iii) A magnetic transition: the dynamo switches on ≈ 350 Ma after the end of the ejections, consistent with the Jack Hills zircons (Tarduno et al., 2025).

With an ejected mass of $2.2\text{--}4.0 \times 10^{22}$ kg per episode and a capture efficiency of ≈ 0.70 , the lunar mass is reconstituted without an external impactor. The isotopic identity and the depletion in metallic iron (reflected in a small lunar core) follow as direct mechanical consequences. The angular momentum budget, however, requires an external post-formation sink of $\approx 3 \times 10^{34}$ J s; the evection resonance (Ćuk & Stewart, 2012; Ćuk et al., 2016) — which precisely requires the high obliquity predicted by the TPT — provides a quantified candidate, whose validation remains required (Limitation L15, Section 18). The distinction between metallic iron and oxidized iron is explicitly treated in Section 14.4.

The central prediction — key to falsifiability — is the existence of one or more seismic interfaces between 200 and 530 km depth, with impedance contrast $|R| \in [0.01, 0.04]$, imminently testable by **Chang’e 7** (South Pole, August 2026), then by **FSS**, **LEMS**, and **Artemis III** (2028–2029). Preliminary support comes from the Chang’e-6 samples (Yue et al., 2026; Xu et al., 2025): norites dated at 4247 ± 5 Ma and high Fe/Mn ratios in deep olivines, consistent with predictions P3 and P6. This retrodictive framework establishes the coherence and physical plausibility of the scenario; its distinction from the giant impact rests entirely on these observational predictions. If they are disproven, the theory must be too — and we hope, if that turns out to be the case, to one day learn what really built the Moon.

Interactive simulation: https://orion4622.github.io/moon-formation-triple-phase-transition/Animation_Formation_de_la_Lune_v2.html

Keywords: origin of the Moon · Triple Phase Transition · chaotic axial oscillation · Hadean proto-Earth · Bingham-Herschel · Coherent Magmatic Torus · generalized instability coefficient · parametric elliptical instability · seismic interface · Hadean dynamo · falsifiable predictions · South Pole–Aitken · Chang’e-6.

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1 Introduction: why a new theory of lunar formation?

1.1 Open observational constraints on lunar formation

Although the formation of the Moon has been studied for several decades, a number of observational constraints remain open, which any formation scenario must explain. Four of them are presented here, and motivate directly the approach developed in this work.

1. **Earth–Moon isotopic identity:** the isotopic compositions of oxygen, titanium, tungsten, and chromium are nearly identical between Earth and the Moon, to better than 5 ppm (Wiechert et al., 2001; Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025). Such similarity places a strong constraint on the origin of lunar material: either it originated from reservoirs already isotopically identical to Earth’s, or a complete homogenization mechanism erased any initial difference between distinct reservoirs. Determining which of these two paths actually occurred remains an open question in lunar cosmochemistry.
2. **Small lunar core:** the Moon has a metallic core of radius ≈ 330 km, significantly smaller than Earth’s core relative to the mass of the host planet (O’Neill, 1991; Jones & Drake, 1986). The reason for this metallic-iron depletion — that is, when and by what mechanism the lunar material lost most of its metallic iron, before or during the Moon’s formation — remains to be clarified.
3. **Crustal dichotomy:** the Moon’s far side is thicker (≈ 54 km) and older than the near side (≈ 34 km) (Wieczorek et al., 2013). The origin of this global-scale hemispheric asymmetry is not established with certainty.
4. **Delay of the terrestrial dynamo:** Earth’s magnetic field is detected in Jack Hills zircons at 4.2 Ga (Tarduno et al., 2025), i.e., ≈ 350 Ma after the estimated formation of the Moon (4.55 Ga). The physical mechanism linking the Moon’s formation to the delayed onset of the terrestrial dynamo remains to be explained.

A recent review highlights the current absence of unambiguous geochemical or isotopic evidence allowing a decisive choice among the various formation scenarios proposed (Sossi et al., 2025). This situation motivates the exploration of approaches grounded in the internal dynamics of the proto-Earth.

1.2 The Triple Phase Transition: an endogenous alternative

This work proposes that the Moon did not form through an external impact, but through an endogenous process within the differentiating proto-Earth itself. The central thesis is as follows.

Hypothesis H0 (central thesis). The Moon formed through a series of *three coupled transitions* — rheological, mechanical, and magnetic — triggered by the single engine

of metallic Fe-Ni segregation toward the forming core. The result is a Moon built layer by layer in two to three episodes of cohesive magmatic ejection, each layer archiving the state of Hadean differentiation at the moment of its accretion.

The mechanism rests on a necessary thermodynamic observation: any Earth-mass planet emerges from accretion in a state of near-total melting of the silicate mantle (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The accretion energy exceeds the total fusion energy by a factor ≈ 121 . The proto-Earth is therefore a rapidly rotating magmatic body ($T_{\text{rot}} \approx 3.5$ h) without a stabilizing satellite.

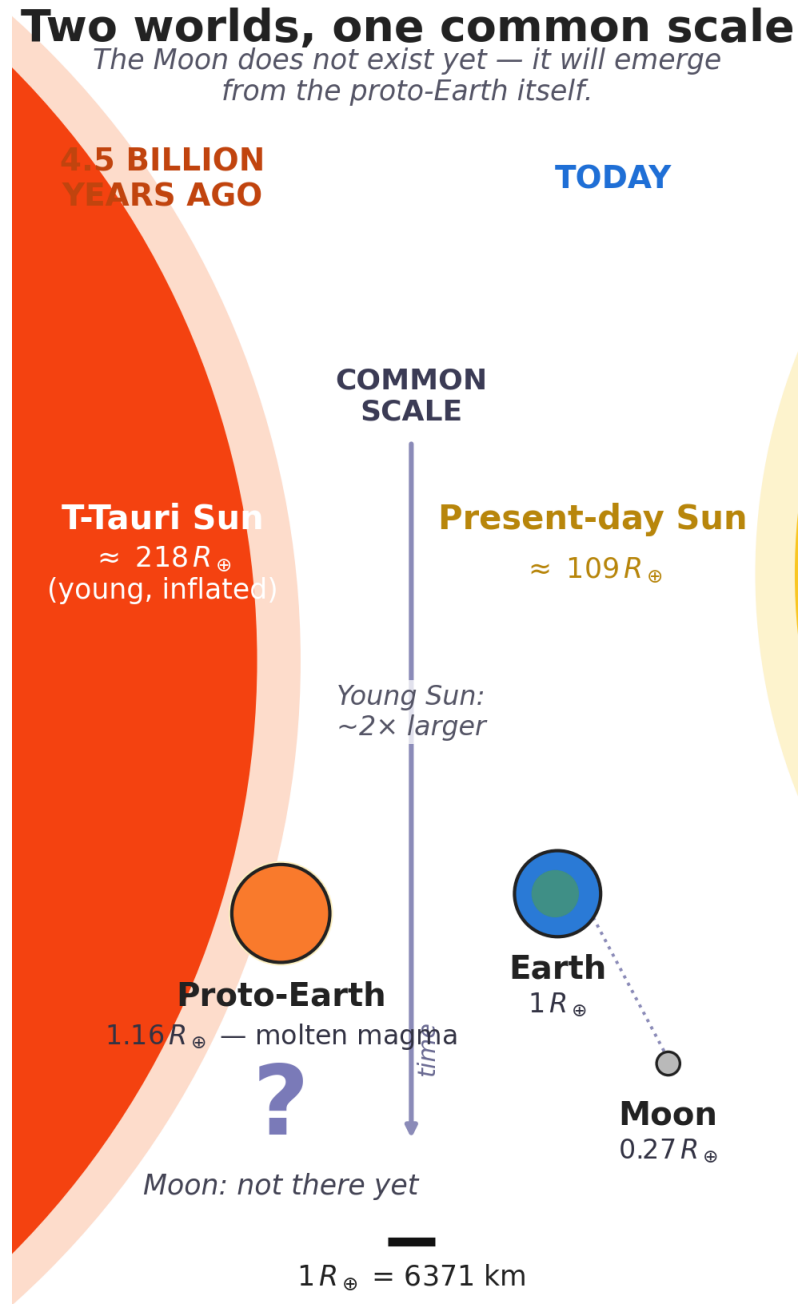
In this rapidly rotating fluid body, the rotation axis undergoes large-amplitude free nutation within the interval $[40^\circ, 70^\circ]$ in the co-rotating frame (Laskar et al., 1993a; Touma & Wisdom, 1993; Li & Batygin, 2014) — not an astronomical obliquity, but an internal oscillation of the fluid body. This oblique tilt is the geometric condition that allows the radial Coriolis force ($f_{\text{rad}} = 2\Omega U \sin \epsilon$) to destabilize the flow rather than confine it.

1.3 Organization of the manuscript

The manuscript is organized as follows. Section 2 presents the Hadean physical context (rotation, obliquity, silicate atmosphere). Section 5 presents the intertropical band as the ejection site. Section 6 answers the Taylor-Proudman objection. Section 7 inventories the forces. Section 8 explicitly classifies all hypotheses. Section 9 presents the three coupled transitions (rheological, mechanical, magnetic) with the full resolution of the instability coefficient. Section 10 presents the mass budget. Section 11 presents angular momentum transport. Section 12 gives the 3-to-4-week chronology. Section 13 presents the internal stratification. Section 14 discusses observational constraints and existing evidence. Section 15 presents the falsifiable predictions. Section 16 presents the validation windows. Section 17 discusses the main objections. Section 18 presents the limitations. Section 19 concludes.

2 The Hadean inferno: physical context

Before diving into the physics of the Triple Phase Transition, it is essential to establish a structural, not merely descriptive, fact about the framework in which it operates: no steady state is physically accessible within it. This is not merely a turbulent period among others in the history of the Solar System — it is a regime in which every process that could, in isolation, bring the system back toward equilibrium is itself disrupted, on a timescale shorter than its own relaxation time, by at least one other independent process. It is this structural absence of rest, and not an inventory of individual circumstances, that this section aims to establish: it directly conditions the reading of Section 3 (the axial oscillation is never damped for lack of a stabilizer) and, further on, that of Section 9.2 (each ejection episode occurs not on a system at rest, but on a system that is already, and continuously, out of equilibrium).



*Suns truncated (arcs); true radius indicated.
 Proto-Earth, Earth and Moon at true relative scale.*

Figure 1: Two worlds, one common scale. On the left, 4.5 billion years ago: the young, inflated T-Tauri Sun ($\approx 218 R_{\oplus}$), and the molten proto-Earth ($1.16 R_{\oplus}$) — the Moon does not exist yet; it will emerge from the proto-Earth itself. On the right, today: the present-day Sun ($\approx 109 R_{\oplus}$), Earth ($1 R_{\oplus}$), and the Moon ($0.27 R_{\oplus}$). Suns truncated (arcs), true radius indicated; proto-Earth, Earth, and Moon at true relative scale.

2.1 The dynamical context of terminal accretion

The Hadean period (4.568–4.4 Ga) superimposes a set of physical processes whose characteristic timescales overlap without any one dominating long enough to impose a stable regime:

- **Continuous accretion of planetesimals:** the protoplanetary disk remains populated by planetesimals and embryos, each major impact occurring every 10^3 – 10^5 years (Agnor et al., 1999; Kokubo & Genda, 2010) — an interval much shorter than the characteristic viscous relaxation time of the fluid body (Section 3), so that the effect of one impact never has time to fade before the next.
- **Rapid rotation:** the proto-Earth rotates with a period ≈ 3.5 h, a regime in which Coriolis and centrifugal effects dominate the internal dynamics and amplify any perturbation rather than damping it.
- **Absence of a tidal stabilizer:** without a Moon, the obliquity is free to evolve over a wide chaotic range, with no restoring mechanism toward an equilibrium value.
- **A young and hyperactive Sun:** T-Tauri-type stars produce flares and coronal mass ejections (CMEs) at frequencies reaching 0.2–0.4 events per day for superflares associated with a confirmed mass ejection, and up to more than one event per day for white-light flares alone, based on direct observations of young solar-type stars (~ 30 – 125 Ma) (Namekata et al., 2026), versus roughly one superflare every few thousand years for the present-day Sun (Feigelson & Montmerle, 1999; Airapetian et al., 2016) — a stochastic forcing whose characteristic timescale is daily, whereas that of planetesimal impacts is millennial and that of free nutation is also of order a day (Section 3): these three clocks never synchronize; they collide.
- **Short-lived radioactivity:** the isotopes ^{26}Al and ^{60}Fe are still active, continuously injecting heat into the mantle (Urey, 1955; Huss et al., 2009), with no thermal resting phase.
- **Dense silicate atmosphere:** an envelope of rock vapor at $T \sim 2\,000$ – $4\,000$ K and $P \sim 10^5$ – 10^7 Pa constitutes a thermal blanket and a lateral confinement agent (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013), but does not by itself stabilize anything: it prolongs the window during which the preceding processes can continue to act.

It is not the list of these processes that matters, but their simultaneity. Taken separately, each could in principle average out over time and allow a quasi-stationary behavior to emerge; taken together, at frequencies that are not commensurable on any common timescale, they structurally prevent such a regime from settling in. The T-Tauri Sun, although it provides one of the most frequent forcings, is therefore not the cause of Hadean chaos: it is merely one contributor among others, whose

combination makes any equilibrium inaccessible. It is within this already, and continuously, out-of-equilibrium context that the ejection episodes themselves will occur (Section 9.2): not as isolated perturbations striking an otherwise calm system, but as additional shocks added to a system that has never, at any point in its active history, truly rested.

2.2 The silicate atmosphere: thermal blanket and lateral confinement

At $t = 0$, the proto-Earth is enveloped in a dense silicate atmosphere — rock vapor, gaseous SiO, Mg, and Fe — at temperatures of 2 000 to 4 000 K and pressures of 10^5 – 10^7 Pa. This cocoon plays two critical roles in the TPT.

First role: thermal insulation. The silicate atmosphere has strong infrared opacity. It reduces surface radiative cooling by a factor of 10^4 to 10^5 (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013). Without this atmosphere, the proto-Earth would have cooled in $\sim 10^4$ years. With it, the melting window extends to 30–50 Ma — exactly the timescale of core formation (Kleine et al., 2002; Yin et al., 2002). This is not an ad hoc hypothesis: it is the physical consequence of a rock-vapor atmosphere at 2 000–4 000 K. The same mechanism operates in magma-ocean models (Elkins-Tanton, 2008; Lebrun et al., 2013).

Second role: lateral confinement. The confining pressure ($P \approx 10^5$ – 10^7 Pa) opposes the lateral expansion of the CMT flow during ejection. It guarantees the cohesion of the ejected sheet. The high speed of sound in this torrid atmosphere ($c_{\text{sound}} \approx 1\,500 \text{ m s}^{-1}$ at $T_{\text{atm}} \approx 3\,000 \text{ K}$) reduces the effective Mach number of the flow, improving cohesion compared to ejection into a vacuum.

Without this atmosphere, the melting window would close in $\sim 10^4$ years. With it, the window extends to 30–50 Ma (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013), which corresponds to the timescale of core formation (Kleine et al., 2002; Yin et al., 2002).

2.3 Continuous planetesimal bombardment

The protoplanetary disk remains populated with planetesimals and embryos throughout the active TPT window (Agnor et al., 1999; Kokubo & Genda, 2010; Chambers, 2001). Each impact: (1) reinjects thermal energy into the mantle, sustaining melting; (2) generates a spherical pressure wave propagating through the entire melt bath, perturbing $U(r)$; (3) delivers an impulsive torque on the spin vector, resetting the axial oscillation.

2.4 Short-lived radioactivity: an internal heat source

The isotopes ^{26}Al (half-life 0.72 Ma) and ^{60}Fe (half-life 2.6 Ma) are still active during the first million years (Urey, 1955; Huss et al., 2009), injecting $\approx 3 \times 10^{29} \text{ J}$ of additional heat into the mantle and contributing to sustained melting throughout the

TPT window.

3 Hadean obliquity $\varepsilon \in [40^\circ, 70^\circ]$: a fundamental geometric clarification

The interval $\varepsilon \in [40^\circ, 70^\circ]$ is not a free parameter. It is constrained by five independent, convergent physical arguments. Before enumerating them, a fundamental geometric clarification is needed.

Geometric clarification. In this work, ε denotes the instantaneous angle between the rotation vector $\mathbf{\Omega}(t)$ and the principal axis of inertia of the Maclaurin spheroid, **defined in the body's co-rotating frame**. It is *not* the astronomical obliquity relative to the ecliptic plane: this concept has no physical meaning for the Hadean proto-Earth, which possesses neither a fixed reference ecliptic nor a Moon. It is a purely internal geometric angle describing the *oscillation* — the free nutation — of the rotation axis relative to the body's own shape.

Definition of the intertropical band. The Hadean intertropical band $|\phi| < 30^\circ$ is defined perpendicular to the *instantaneous* rotation axis $\mathbf{\Omega}(t)$, and not relative to a fixed geographic reference frame. The plane $\phi = 0$ is thus the body's *instantaneous dynamical equator*, which oscillates with the free nutation of $\mathbf{\Omega}(t)$ — it is not a fixed geographic equator. The coordinate ϕ is the oblique latitude measured from this instantaneous equator. This is the plane where the radial Coriolis force $f_{\text{rad}} = 2\Omega U \sin \varepsilon$ is maximal and where the effective gravity $g_{\text{eff}}(\phi)$ is minimal (Section 4.3). The band is therefore a dynamical structure tied to the instantaneous equator, not a fixed latitude.

It follows directly from the absence of any external reference plane in the Solar System at this epoch — neither ecliptic, nor stable lunar orbit, nor fixed solar axis — that the very notion of an “intertropical” band is not geographic but purely relative, defined by this internal axis rather than by a solar or ecliptic reference frame that does not yet exist. Throughout this work, terms evoking ordinary planetary geography (“intertropical,” “equator,” “latitude”) should be read as a formal analogy with the geometry of a rotating spheroid, and not as a reference to an astronomy that has not yet come into being at this epoch.

Argument 1 — Absence of a tidal stabilizer (fundamental). The proto-Earth has no Moon. Without a massive satellite, there is no lunar tidal torque to damp the precession of the spin axis and keep it near a stable equilibrium. This argument is logically prior to all the others: it is precisely the formation of the Moon that we seek to explain, so invoking it as a stabilizer would be circular. The absence of a tidal stabilizer is therefore a necessary, not contingent, condition of the Hadean proto-Earth.

Argument 2 — Laskar et al. (1993): a chaotic zone without a satellite. Laskar et al.

(1993a) showed that without its Moon, Earth’s obliquity would evolve chaotically over a zone extending from nearly 0° to $\approx 85^\circ$. Although this result strictly applies to a present-day Earth rotating in ≈ 24 h, it establishes the general principle: *the Moon is the stabilizer, and without it, large obliquity excursions are the norm*. For the Hadean proto-Earth, this principle applies *a fortiori*.

Argument 3 — Rapid rotation at $T_{\text{rot}} = 3.5$ h. The precession constant satisfies $\alpha_{\text{prec}} \propto \omega$. At $T_{\text{rot}} = 3.5$ h, the precession frequency is $\approx 7\times$ higher than at 24 h, pushing the secular spin-orbit resonances identified by Laskar et al. (1993a) out of reach. Li & Batygin (2014), working on a rapidly rotating Moonless Earth, found that the obliquity remains chaotic but *diffuses slowly* over a wide range. Rapid rotation does not stabilize the axis — it simply changes the nature of the chaos, from resonant to diffusive.

Argument 4 — Continuous bombardment and stochastic chaos. Every major impact during terminal accretion (Agnor et al., 1999) abruptly resets the spin vector and the shape of the fluid body. For a low-viscosity fluid body ($\eta \sim 10^{1-3}$ Pa s), the viscous relaxation time toward axial alignment, of the form $\tau \sim \eta / (\rho g R_{\text{eq,proto}})$ (Goldreich & Mitchell, 2010), is extremely short ($\ll 1$ year) compared to the interval between major impacts (10^3 – 10^5 years): $\tau_{\text{relax}} \ll \tau_{\text{pert}}$. The system therefore realigns between each impact and spends more than 99.9% of the time aligned. Obliquity is therefore not “continuously maintained” in the sense of a permanent hold at $\varepsilon > 45^\circ$.

What the impacts do ensure, however, is that the system never reaches a steady state. Each impact resets the initial conditions of the free nutation, and the continuous stochastic forcing (CMEs, bombardment, late accretion) is distributed over timescales that are not commensurable. Obliquity is therefore not stabilized; it is continually *forced* toward new values. The interval $\varepsilon \in [40^\circ, 70^\circ]$ is not maintained by a slow relaxation mechanism, but by the absence of a stabilizer and the persistence of stochastic forcing that prevents the system from freezing. It is this absence of rest — and not an active maintenance — that makes large obliquity excursions accessible.

Argument 5 — T-Tauri CME torques. As described in Section 7.4, the impulsive torques of CMEs act on timescales of days to weeks, comparable to τ_{pert} , and constitute a continuous stochastic source of obliquity perturbation, in addition to impacts (Feigelson & Montmerle, 1999; Airapetian et al., 2016).

The interval $\varepsilon \in [40^\circ, 70^\circ]$ is not an ad hoc hypothesis. It is the predictable consequence of the dynamics of a rapidly rotating fluid body, lacking a tidal stabilizer, subjected to continuous stochastic perturbations. The simulations of Laskar et al. (1993a) for a Moonless Earth, and of Li & Batygin (2014) for a rapidly rotating Moonless Earth, converge on this interval — the same one adopted by Laskar et al. (1993a) and Touma & Wisdom (1993). In this oblique geometry, the Coriolis force decomposes into radial and horizontal components (Section 7.3). The radial component $f_{\text{rad}} = 2\Omega \sin \varepsilon \cdot U$ is maximal for $\varepsilon \in [40^\circ, 70^\circ]$.

Section summary

Five convergent arguments support $\varepsilon \in [40^\circ, 70^\circ]$. The absence of a tidal stabilizer is logically necessary. Arguments 2 through 5 provide independent physical justification for a wide and sustained misalignment. The interval $[40^\circ, 70^\circ]$ is adopted as the physically motivated Hadean interval; the mechanism operates for any $\varepsilon > 45^\circ$ (Section 7.3). The intertropical band $|\phi| < 30^\circ$ is, throughout this work, tied to the instantaneous dynamical equator $\Omega(t)$, not to a fixed geographic reference frame.

3.1 Relaxation time and rheological reinforcement

The viscous relaxation time toward axial alignment for an isolated fluid body, $\tau_{\text{relax}} \sim \eta / (\rho g R_{\text{eq,proto}})$ (Goldreich & Mitchell, 2010), is very short ($\ll 1$ year) at low viscosity, as established in Section 3. The Bingham-Herschel rheology locally slows this relaxation. The rheological response time to a stress perturbation is:

$$\tau_{\text{BH}} \sim \frac{\tau_y}{\mu \dot{\gamma}_{\text{typ}}} \approx 10^2\text{--}10^4 \text{ s}, \quad (1)$$

with $\tau_y \sim 10^2\text{--}10^4$ Pa, $\mu \sim 1\text{--}100$ Pa s, and $\dot{\gamma}_{\text{typ}} \sim 1 \text{ s}^{-1}$ (Giordano et al., 2008), where μ is the dynamic viscosity and $\dot{\gamma}_{\text{typ}}$ the typical shear rate. This time remains much shorter than the interval between impacts τ_{pert} , meaning that mass redistribution is locally frozen by the yield stress between perturbations, delaying local shape relaxation without affecting the conclusion of Argument 4: it is the absence of rest, due to the frequency of perturbations and the absence of a stabilizer, that keeps the system out of equilibrium.

4 Initial state of the proto-Earth**4.1 Initial rotation period: $T_{\text{rot}} \approx 3.5$ h**

The initial rotation period of the proto-Earth is not a free parameter: it is directly constrained by the literature on terrestrial-planet formation at $T_{\text{rot}} \approx 3\text{--}5$ h.

1. T-Tauri magnetic torque. The young, rapidly rotating Sun (≈ 3 days, Bouvier et al. 2014) couples magnetically to the inner protoplanetary disk (Königl, 1991; Shu et al., 1994), transferring angular momentum to matter on the inner orbit and accelerating the proto-Earth.

2. N-body simulations of terrestrial planet formation. N-body simulations show that the final spin periods of Earth-mass planets lie in the range 3–5 h (Agnor et al., 1999; Kokubo & Genda, 2010; Chambers, 2013). Agnor et al. (1999) find that protoplanetary embryos rotate rapidly during the late stages of accretion, at their first collision, suggesting that the proto-Earth may have rotated rapidly before any

subsequent major impact. Kokubo & Genda (2010), using N-body simulations of protoplanet formation under realistic accretion conditions, find final rotation periods of 3–5 h for Earth-mass bodies, consistent with Chambers (2013). The value $T_{\text{rot}} \approx 3.5$ h adopted here lies within this independently established interval, and is of the same order of magnitude as the rapid pre-impact rotations considered in the giant-impact models of Čuk & Stewart (2012) (2.3–2.7 h) and Canup (2012) (a pre-impact rotation of 3 h in one of the simulated cases) — although none of these specific values is identical to 3.5 h, and the TPT does not rely on the giant-impact hypothesis itself either.

Why rapid rotation requires an acceleration mechanism. Dones & Tremaine (1993) show analytically that orderly accretion from a uniform or slowly varying disk of small bodies cannot produce rotation as rapid as that of Earth or Mars, regardless of the disk’s velocity dispersion: the prograde and retrograde contributions of individual planetesimals nearly cancel by symmetry. The rapid spin rates of terrestrial planets are therefore most naturally explained by one or a few collisions between large bodies late in accretion — a requirement that the TPT satisfies through the continuous planetesimal bombardment of Section 2, independently of whether a single dominant impactor is invoked (as in the giant-impact hypothesis). Consistent with this, Kokubo & Ida (2007), using N-body simulations of the embryo-merging phase of accretion, find that planetary spin angular velocities follow a Maxwellian distribution with substantial dispersion: there is no single preferred rotation period predicted by accretion physics alone, but only a broad statistical range within which $T_{\text{rot}} \approx 3.5$ h is a plausible value.

3. Skater effect: Fe-Ni segregation. The migration of Fe-Ni toward the forming core reduces the moment of inertia as segregation proceeds. With $I_f / I_0 \approx 0.82$ (Rubie et al., 2015), this contributes an additional acceleration of about 18% to rotation over the TPT window, on top of the rotation speed already acquired by the proto-Earth during accretion. This effect is qualitative and contributory: it is not used here to derive $T_{\text{rot}} \approx 3.5$ h from an assumed pre-segregation value, since no such value is independently constrained.

4.2 Why rapid rotation: a dedicated subsection

The rapid rotation of the proto-Earth ($T_{\text{rot}} \approx 3.5$ h) is not a decorative hypothesis. It is required by the physics of accretion and by the TPT mechanism itself. This subsection summarizes the five independent reasons that constrain this value.

1. **Orderly accretion is not enough.** Dones & Tremaine (1993) show analytically that symmetric accretion from a disk of small bodies cannot produce rapid rotation: the prograde and retrograde contributions nearly cancel. A rapid spin requires an acceleration mechanism late in accretion.
2. **N-body simulations impose 3–5 h.** Kokubo & Ida (2007) find that the an-

gular velocities of terrestrial planets follow a Maxwellian distribution with substantial dispersion. 3.5 h is a plausible value within this interval.

3. **Fe-Ni segregation accelerates rotation.** The migration of metallic iron toward the core reduces the moment of inertia by about 18% (Rubie et al., 2015), which accelerates rotation through the skater effect.
4. **The ejection threshold requires rapid rotation.** The condition $\Omega > \Omega_{\text{crit}}$ (Section 9.2) is only reachable for $T_{\text{rot}} \lesssim 4$ h. A slower proto-Earth could never eject matter.
5. **The proto-Earth is not a synestia.** The corotation limit (CoRoL) of Lock & Stewart (2017) is ≈ 1.8 h for the proto-Earth parameters. At 3.5 h, Ω reaches only $\approx 50\%$ of this limit: the body remains a coherently rotating spheroid, not a distended structure.

On the initial rotation period

The value $T_{\text{rot}} \approx 3.5$ h is directly constrained by the literature on terrestrial-planet formation (Agnor et al. 1999; Kokubo & Genda 2010; Chambers 2013; 3–5 h), is of the same order of magnitude as the rapid rotations considered in the giant-impact models (Ćuk & Stewart, 2012; Canup, 2012) without depending on the giant-impact hypothesis, and is consistent with the theoretical requirement that rapid terrestrial spin requires an acceleration mechanism late in accretion (Dones & Tremaine, 1993), within the broad statistical range found for planetary spin in N-body simulations (Kokubo & Ida, 2007). It is not derived from an assumed initial period via a contraction argument. The Coriolis compensation parameter b' , a simple ratio between the critical ejection velocity U_{crit} and the CMT's geostrophic equilibrium velocity, provides an independent cross-check: Fe-Ni segregation increases b' , ejections decrease it, and the equilibrium point $b' = 1$ is a stable attractor that naturally selects $T_{\text{rot}} \approx 3.5$ h and $\varepsilon \approx 55^\circ$. The full derivation is given in Appendix B.

4.3 Geometry of the proto-Earth: the Maclaurin spheroid

A rapidly rotating fluid mass adopts the shape of a Maclaurin spheroid (Chandrasekhar, 1969). The equatorial radius is $R_{\text{eq,proto}} \approx 7.4$ Mm, combining reduced density (+4%), thermal expansion (+7.5%), and equatorial bulging (+7.5%). The dynamical flattening, given by $J_2 = q/[2\Phi(\bar{C}) - 3]$ where $q = \Omega^2 R_{\text{eq,proto}}^3 / (GM_\oplus)$ is the rotation parameter and $\bar{C}$ the normalized axial moment of inertia (Murray & Dermott, 1999), reduces to $J_2 \approx q/2$ for a strictly homogeneous body ($\bar{C} = 0.4$):

$$J_2 \approx \frac{1}{2} \cdot \frac{\Omega^2 R_{\text{eq,proto}}^3}{GM_\oplus} \approx 0.13, \quad (2)$$

a value consistent with the interval $J_2 \sim 0.10\text{--}0.13$ expected for a body close to homogeneity (Murray & Dermott, 1999), giving an equatorial bulge $\delta R \approx 600\text{--}650$ km. Effective gravity varies with latitude ϕ (measured from the instantaneous dynamical equator, as defined in Section 3) according to:

$$g_{\text{eff}}(\phi) = g_{\text{eff}}^{(0)} \left(1 + \frac{3}{2} J_2 \sin^2 \phi \right) - \Omega^2 r(\phi) \cos^2 \phi, \quad (3)$$

where $r(\phi)$ is the local radius. Effective gravity is minimal within the band $|\phi| < 30^\circ$: this is the CMT's geometric attractor.

The free (Euler) nutation period, $\omega_E = \Omega(C - A)/A$ where C and A are the spheroid's polar and equatorial moments of inertia (Murray & Dermott, 1999), with $(C - A)/A \equiv f \approx 0.075$ for a body close to homogeneity at the dynamical flattening factor J_2 adopted here (Eq. 2), is:

$$\omega_E = \Omega f \approx 3.74 \times 10^{-5} \text{ rad s}^{-1}, \quad T_{\text{nut}} \approx 46.7 \text{ h}. \quad (4)$$

This frequency plays a central role in the parametric elliptical instability (Section 7.6).

Stability of the Maclaurin spheroid at this rotation rate. A rapidly rotating self-gravitating fluid body is not necessarily constrained to adopt the axisymmetric Maclaurin shape: beyond a critical rotation rate, the Maclaurin sequence bifurcates toward the triaxial Jacobi sequence. The bifurcation criterion compares two characteristic timescales: the self-gravitation timescale, $t_g \sim 1/\sqrt{G\bar{\rho}}$, and the rotation period, $t_{\text{rot}} = 2\pi/\Omega$; their ratio defines the dimensionless number $\Omega^2/(\pi G\bar{\rho})$. This criterion is analytically derived by Chandrasekhar (1969) (chapter 4, full demonstration; see also chapters 3 to 5 for Maclaurin spheroids and chapters 6 and 7 for Jacobi ellipsoids and the bifurcation itself) as the point at which the family of Jacobi ellipsoids branches off from the Maclaurin sequence — and not as an empirical approximation — and is presented in historical perspective by Jeans (1929), taken up in the stellar context by Tassoul (1978), the theory being formally identical for any self-gravitating fluid; Lamb (1932) gives the classical hydrodynamic treatment, and Durand (1964) the ellipsoidal integrals on which Chandrasekhar's demonstration rests. This bifurcation occurs at $\Omega^2/(\pi G\bar{\rho}) \approx 0.374$, with Maclaurin spheroids becoming truly unstable only beyond $\Omega^2/(\pi G\bar{\rho}) \approx 0.440$.

The ρ in this criterion is the uniform density of the homogeneous fluid in the theoretical model. Its application to a real planetary body substitutes for this uniform density the mean density $\bar{\rho} = M/(\frac{4}{3}\pi R^3)$, an approximation explicitly discussed by Chandrasekhar (1969) (chapters 1 and 5) and Tassoul (1978) (chapter 2). For the Hadean proto-Earth, $\bar{\rho}$ is not directly measurable: it depends on the degree of differentiation already reached, which is itself not precisely known. Available estimates place the mean density of a homogeneous Earth of the same mass, prior to complete core segregation, around $4\,100\text{--}4\,500 \text{ kg m}^{-3}$, versus $5\,510 \text{ kg m}^{-3}$ for

today’s fully differentiated Earth (Rubie et al., 2015; Trønnes et al., 2019; Rubie et al., 2016; Hirose et al., 2021). These values are themselves derived from mineralogical models (chondritic or pyrolitic composition, equations of state) rather than direct measurement, and therefore constitute a modeling range rather than a physical constant.

For the proto-Earth parameters adopted here ($R_{\text{eq,proto}} \approx 7.4$ Mm, mean density $\bar{\rho} \approx 4300 \text{ kg m}^{-3}$, the median value of this interval, for a still weakly differentiated body), $T_{\text{rot}} \approx 3.5$ h gives $\Omega^2 / (\pi G \bar{\rho}) \approx 0.276$ — below the Jacobi bifurcation point (0.374), with a margin of about 26%. At this density, the bifurcation threshold itself corresponds to $\Omega_c = \sqrt{0.374 \pi G \bar{\rho}} \approx 5.8 \times 10^{-4} \text{ rad s}^{-1}$, i.e., $T_{\text{rot}} \approx 3.0$ h, and the strict dynamical instability threshold at $T_{\text{rot}} \approx 2.8$ h. The axisymmetric Maclaurin geometry adopted throughout this work is therefore consistent with the TPT’s rotation rate, and does not require resorting to a triaxial or otherwise more complex equilibrium figure; the available margin nonetheless remains sensitive to the exact value of $\bar{\rho}$ adopted, which is not known with a precision better than the modeling range cited above.

Distinction from the synestia regime. Lock & Stewart (2017) define the corotation limit (CoRoL) as the rotation speed at which a body’s equatorial velocity equals the local Keplerian orbital velocity; bodies exceeding this limit form a distended, partially vaporized structure they call a synestia, rather than a simply rigidly rotating body. For the proto-Earth equatorial radius derived above ($R_{\text{eq,proto}} \approx 7.4$ Mm), the corotation limit corresponds to a period of about 1.8 h. At $T_{\text{rot}} \approx 3.5$ h, Ω reaches only $\approx 50\%$ of this limit: the proto-Earth described by the TPT thus lies well within the regime of a coherently rigidly rotating body, not a synestia.

4.4 Near-total melting: a necessary thermodynamic consequence

The formation of an Earth-mass planet releases a considerable amount of gravitational energy. In the homogeneous-sphere approximation (Safronov, 1969), the total accretion energy is:

$$E_{\text{grav}} = \frac{3 G M_{\oplus}^2}{5 R_{\text{eq,proto}}} \approx 1.93 \times 10^{32} \text{ J}, \quad (5)$$

where G is the gravitational constant, $M_{\oplus}$ Earth’s mass, and $R_{\text{eq,proto}} \approx 7.4$ Mm the equatorial radius of the undifferentiated proto-Earth.

Equation (5) is a thermodynamic bound, not a model of instantaneous formation. It computes the *total* gravitational energy released during accretion, independent of its duration. This is the standard approach in the literature (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The manuscript explicitly cites Kleine et al. (2002); Yin et al. (2002) as evidence that accretion lasted ~ 30 Ma.

The energy required to melt the entire silicate mantle (Stixrude & Lithgow-Bertelloni,

2014):

$$E_{\text{fus}} \approx M_{\text{mantle}} \times L_{\text{fus}} \approx 4 \times 10^{24} \text{ kg} \times 4 \times 10^5 \text{ J kg}^{-1} \approx 1.6 \times 10^{30} \text{ J}, \quad (6)$$

where M_{mantle} is the mass of the silicate mantle and L_{fus} the specific latent heat of fusion.

The resulting energy ratio is:

$$E_{\text{grav}} / E_{\text{fus}} \approx 121. \quad (7)$$

This precise value follows from the parameters adopted here ($R_{\text{eq,proto}} \approx 7.4 \text{ Mm}$, $M_{\text{mantle}} \approx 4 \times 10^{24} \text{ kg}$, $L_{\text{fus}} \approx 4 \times 10^5 \text{ J kg}^{-1}$), explicitly stated above; it is not a universal Earth constant. Other, equally defensible, choices of parameters in the literature — a radius close to Earth’s current radius, a smaller mantle mass, or a higher latent heat — give different numerical values, typically between ≈ 80 and ≈ 190 depending on the assumptions adopted. What remains robust across this entire range, and what constitutes the consensus result in the literature (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015), is the qualitative conclusion: accretion energy exceeds the fusion energy of the silicate mantle by more than two orders of magnitude, making the near-total melting of the proto-Earth a robust thermodynamic consequence rather than an outcome contingent on a particular choice of parameters.

Why melting was sustained despite radiative cooling

Accretion lasted $\sim 30 \text{ Ma}$ (Kleine et al., 2002; Yin et al., 2002). During this period, radiative cooling occurred. Four independent mechanisms explain why melting was sustained despite this cooling.

Mechanism 1 — The silicate atmosphere as a thermal blanket. The dense silicate atmosphere ($P \approx 10^5\text{--}10^7 \text{ Pa}$, $T \approx 2\,000\text{--}4\,000 \text{ K}$) has strong infrared opacity. It reduces surface radiative cooling by a factor of 10^4 to 10^5 (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013). The effective cooling time goes from $\sim 10^4$ years to 30–50 Ma.

Mechanism 2 — Additional heat sources. Four heat sources extend melting beyond the initial accretion energy:

- **Fe-Ni segregation heat:** $\approx 3 \times 10^{30} \text{ J}$ (Flasar & Birch, 1973; Rubie et al., 2003).
- **Short-lived radioactivity:** ^{26}Al (half-life 0.72 Ma) and ^{60}Fe (half-life 2.6 Ma) inject $\approx 3 \times 10^{29} \text{ J}$ (Urey, 1955; Huss et al., 2009).
- **Continuous planetesimal bombardment:** reinjection of thermal energy (Agnor et al., 1999; Kokubo & Genda, 2010).
- **T-Tauri irradiation:** keeps surface temperatures above 2000 K (Feigelson &

Montmerle, 1999; Airapetian et al., 2016).

Mechanism 3 — The TPT does not require melting of the entire mantle. The Triple Phase Transition requires melting only within the Hadean intertropical band $|\phi| < 30^\circ$ (upper and mid mantle). Factor 121 thermodynamically guarantees this partial melting, even if the deep mantle was not entirely molten (Nomura et al., 2011).

Mechanism 4 — The active TPT window is 30–50 Ma. The TPT occurs *during* the 30–50 Ma window in which melting is sustained by the mechanisms above. Core formation over ~ 30 Ma (Kleine et al., 2002; Yin et al., 2002) follows this window, or is simultaneous with it — this is not a contradiction.

Near-total melting is the mandatory starting point. The objection based on the ~ 30 Ma accretion timescale does not contradict the mechanism: the TPT occurs *during* the melting window sustained by the silicate atmosphere, the additional heat sources, and continuous bombardment.

Hypothesis H1 — Total volumetric melting. At the onset of Fe-Ni segregation, the entire silicate mantle lies above the liquidus, with no stable internal solid interface during the active window 4.568–4.4 Ga. This is a self-gravitating fluid volume, not a magma ocean resting on a substrate: the dynamics, rheology, and response to perturbations are physically distinct.

Caveat: Nomura et al. (2011) suggest that the deep lower mantle may not have been entirely molten, owing to the rise of the solidus with pressure. The TPT only requires melting within the Hadean intertropical band $|\phi| < 30^\circ$ (upper and mid mantle), which factor 121 thermodynamically guarantees. The adiabatic temperature profile (Appendix A) confirms melting throughout the mantle.

5 The intertropical band $|\phi| < 30^\circ$: the ejection site

A fundamental and often misunderstood point: the CMT is not an equatorial structure. It is a coherent zonal band extending over $\pm 30^\circ$ of oblique latitude — about a third of the proto-Earth’s surface. This extent results from three independent, convergent mechanisms.

As established in Section 3, the band $|\phi| < 30^\circ$ is defined perpendicular to the instantaneous rotation axis $\Omega(t)$, and not relative to a fixed geographic reference frame. This is the plane where the radial Coriolis force is maximal and where effective gravity is minimal.

Mechanism 1 — Maclaurin geometry. Effective gravity is minimal within the band $|\phi| < 30^\circ$ (Eq. 3). Molten matter accumulates there through centrifugal effects, independent of any other argument. This is a consequence of the ellipsoidal shape imposed by rapid rotation.

Mechanism 2 — The geostrophic inverse cascade. The CMT's Rossby number, with $U_{\text{turb}} \approx 0.8 \text{ km s}^{-1}$ (Eq. 19), is:

$$\text{Ro} = \frac{U_{\text{turb}}}{2\Omega R_{\text{eq,proto}}} \approx 0.108 \ll 1, \quad (8)$$

In this regime, turbulent energy cascades toward large scales (inverse cascade) (Rhines, 1975). The Rhines scale:

$$L_\beta = \sqrt{\frac{U_{\text{turb}}}{\beta}}, \quad \beta = \frac{2\Omega \cos \varepsilon}{R_{\text{eq,proto}}}, \quad (9)$$

gives $L_\beta \approx 0.47 R_\oplus$ — corresponding to the harmonic mode $\ell = 1, m = 1$: the largest coherent structure possible on a sphere. The CMT is the statistical attractor of this cascade (Rhines, 1975; Sukoriansky et al., 2007).

Mechanism 3 — The radial Coriolis force. $f_{\text{rad}} = 2\Omega U \sin \varepsilon$ concentrates the flow within the oblique intertropical band and keeps it coherent there.

Section summary

These three mechanisms — Maclaurin geometry, the geostrophic inverse cascade, and the radial Coriolis force — are independent and converge on the same band $|\phi| < 30^\circ$. Their simultaneous convergence constitutes an argument for the structural robustness of the theory.

6 The CMT as a purely toroidal flow: answer to the Taylor-Proudman objection

Any geophysical fluid dynamicist confronted with the CMT will naturally raise the following objection: in a rapidly rotating fluid ($\text{Ro} \ll 1$), the Taylor-Proudman theorem imposes $(\Omega \cdot \nabla)\mathbf{u} = 0$, producing columnar structures parallel to Ω , rather than a single oblique band $|\phi| < 30^\circ$. The objection is legitimate and deserves a precise answer.

6.1 The poloidal/toroidal decomposition

Any velocity field within a spherical shell decomposes into a poloidal part (radial and meridional components) and a toroidal part (purely azimuthal component):

$$\mathbf{u} = \mathbf{u}_{\text{pol}} + \mathbf{u}_{\text{tor}}, \quad \mathbf{u}_{\text{tor}} = \nabla \times (\psi \hat{\mathbf{r}}). \quad (10)$$

The Taylor-Proudman theorem, in its classical form, constrains the poloidal field (which carries vorticity along Ω) and produces Taylor columns. It does not constrain the purely toroidal field.

For a purely azimuthal flow independent of longitude ($u_r = u_\theta = 0$ on average, $u_\phi = u_\phi(r, \phi)$), the Taylor-Proudman constraint is automatically satisfied:

$$(\mathbf{\Omega} \cdot \nabla) u_{\text{tor}} = 0 \quad (11)$$

for any toroidal field, with no additional condition. The balance is cyclostrophic, not geostrophic: the centrifugal pressure gradient balances the Coriolis force within the azimuthal plane.

6.2 Why curved columns do not extend beyond the active band

In an obliquely rotating spherical shell ($\varepsilon \neq 0$), Taylor columns are curved along the projected field lines of $\mathbf{\Omega}$. They tend to channel the flow away from its source — unless a cutoff mechanism stops them.

This is precisely the role of the yield stress τ_y . In a Bingham-Herschel fluid, the flow only propagates in regions where the stress exceeds τ_y . As soon as the columns penetrate a region where $\tau < \tau_y$, they are truncated: the magma solidifies there (Frigaard & Nouar, 2017; Balmforth & Craster, 2001). The band $|\phi| < 30^\circ$ is precisely the region where effective gravity is minimal (Maclaurin) and where the Coriolis stress is maximal ($\varepsilon \approx 55^\circ$): it is there, and only there, that $\tau > \tau_y$. Curved columns originating in the active band are thus naturally truncated at its boundaries.

Section summary

The CMT is a purely toroidal flow. The Taylor-Proudman objection does not apply. What remains to be demonstrated numerically — identified here as Limitation L1 (Section 18) — is whether a single torus forms within the band $|\phi| < 30^\circ$ rather than several alternating bands. This is a question for simulation, not a theoretical inconsistency.

7 Complete inventory of forces

All forces, projections, and equations are expressed in the instantaneous co-rotating Hadean frame of the body, with ε defined as in Section 3.

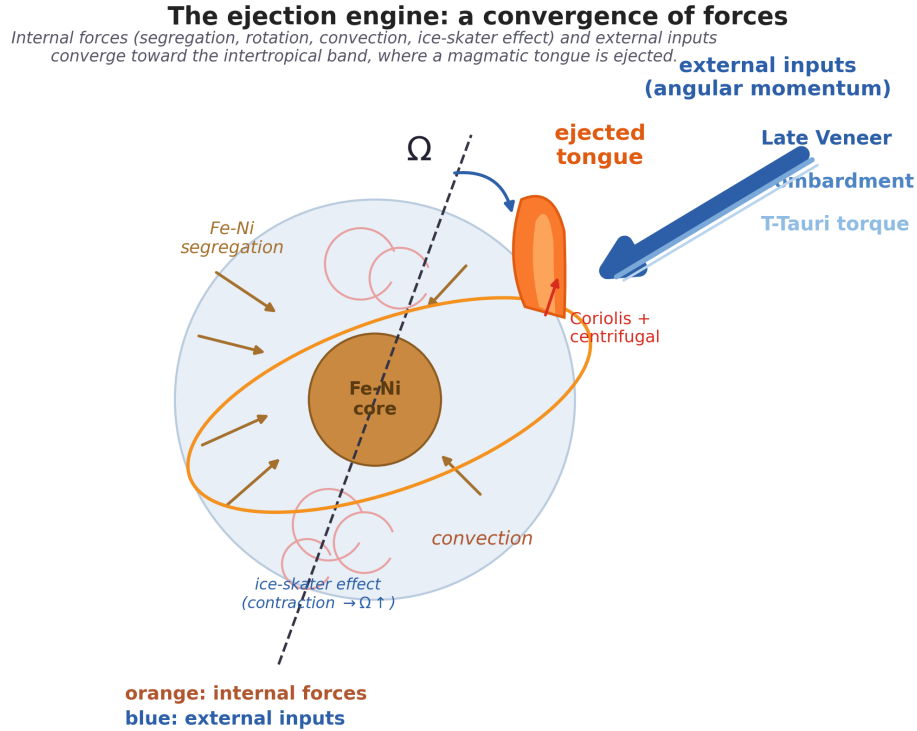


Figure 2: The ejection engine: a convergence of forces. Internal forces (Fe-Ni segregation toward the core, rotation Ω , convection, skater effect: contraction $\rightarrow \Omega \uparrow$) and external angular momentum inputs (Late Veneer, bombardment, T-Tauri torque) converge toward the intertropical band, where a magmatic tongue is ejected under the combined effect of the Coriolis and centrifugal forces. Orange: internal forces; blue: external inputs.

7.1 Complete force inventory: summary table

The CMT is a toroidal flow within a rapidly rotating fluid body, subject to a set of forces. Table 1 summarizes all the forces, their physical role, and their mathematical expression.

Table 1: Complete inventory of the forces acting on the CMT.

Force	Role	Expression
Effective gravity	Radial confinement	$g_{\text{eff}}(\phi) = g_{\text{eff}}^{(0)}(1 + \frac{3}{2}J_2 \sin^2 \phi) - \Omega^2 r(\phi) \cos^2 \phi$
Centrifugal force	Lateral ejection, flattening	$F_{\text{cent}} = \Omega^2 r(\phi) \cos^2 \phi$
Radial Coriolis	Destabilization, outward ejection	$F_{\text{cor,rad}} = 2\Omega U \sin \varepsilon$
Horizontal Coriolis	Lateral confinement	$F_{\text{cor,hor}} = 2\Omega U \cos \varepsilon$
Taylor-Proudman	Vertical columns, toroidal flow	$(\boldsymbol{\Omega} \cdot \nabla) \mathbf{u}_{\text{pol}} = 0; \mathbf{u}_{\text{tor}} \text{ unconstrained}$
Planetesimal bombardment	Stochastic perturbation, thermal reinjection	$\delta \boldsymbol{\Omega} = \sum_i \delta \boldsymbol{\Omega}_i, \delta U = \sum_i \delta U_i$
T-Tauri (CME)	Stochastic perturbation, thermal maintenance	$F_{\text{CME}}(t) = \sum_j F_{\text{CME},j} \delta(t - t_j)$
Solar tide	Low-frequency forcing on obliquity	$F_{\text{tide,sol}} \sim 10^{-4} g_0 \cos(\omega_{\text{sol}} t)$
Proto-lunar tide	Mathieu resonance (episode 2+)	$F_{\text{tide,moon}}(t) = A_m^{(k)}(t) \cos(n_{\text{Moon}} t)$
Euler acceleration	Axial nutation	$\mathbf{a}_{\text{Euler}} = \dot{\boldsymbol{\Omega}} \times \mathbf{r}$
Atmospheric pressure	Lateral confinement	$P_{\text{atm}} \sim 10^5 - 10^7 \text{ Pa}$
BH yield stress	Plasticity threshold	$\tau = \tau_y + k\dot{\gamma}^n, \tau > \tau_y$
Viscous stress	Dissipation	$\tau_{\text{visc}} = \eta \nabla^2 \mathbf{u}$

7.2 Oscillation as a direct force source

The axial oscillation implies $\dot{\boldsymbol{\Omega}} \neq \mathbf{0}$. Poincaré (1910) showed that this generates a Euler acceleration on every fluid particle:

$$\mathbf{a}_{\text{Euler}} = \dot{\boldsymbol{\Omega}} \times \mathbf{r}, \quad (12)$$

where $\mathbf{r}$ is the position vector. The radial projection:

$$\mathbf{a}_{\text{Euler}} \cdot \hat{\mathbf{r}} = |\dot{\boldsymbol{\Omega}}| r(\phi) \sin(\varepsilon + \phi), \quad (13)$$

is maximal within the band $|\phi| < 30^\circ$ where $r(\phi)$ is largest.

7.3 Radial Coriolis force and geometric threshold

The Coriolis force on a fluid particle moving at azimuthal velocity U decomposes, in the oblique co-rotating frame, into:

$$f_{\text{rad}} = 2 \Omega \sin \varepsilon \cdot U \quad (\text{radial, outward, destabilizing}), \quad (14)$$

$$f_{\text{horiz}} = 2 \Omega \cos \varepsilon \cdot U \quad (\text{horizontal, confining}), \quad (15)$$

with the ratio:

$$\frac{f_{\text{rad}}}{f_{\text{horiz}}} = \tan \varepsilon. \quad (16)$$

Geometric threshold $\varepsilon = 45^\circ$ — necessary condition for radial ejection. For $\varepsilon > 45^\circ$: $\tan \varepsilon > 1$, the destabilizing radial component dominates the confining horizontal component. For $\varepsilon < 45^\circ$: the horizontal component dominates and the Taylor-Proudman constraint inhibits radial ejection. This threshold follows purely from Coriolis kinematics: no adjustable parameter is involved. At $\varepsilon = 55^\circ$: $f_{\text{rad}}/f_{\text{tot}} = \sin 55^\circ \approx 0.82$, i.e., 82% of the total Coriolis force directed radially outward.

7.4 The T-Tauri Sun: catalyst and perturber

The Sun in its T-Tauri phase is not a driving force on the same footing as gravity or the Coriolis force — it does not act continuously on every fluid parcel of the CMT. It nonetheless plays three essential roles within the TPT window.

(1) Rheological modulator. T-Tauri UV/X-ray radiation keeps T_{surf} within the range 1 800–4 200 K (Stefan-Boltzmann law, albedo $A_0 \approx 0.1$), which keeps the yield stress τ_y of the silicate magma within the range 10^2 – 10^4 Pa (Giordano et al., 2008) — the optimal window for CMT cohesion. Below $T^* \approx 1 800$ K, τ_y grows to 5.5×10^3 – 8.0×10^4 Pa (Eq. 25) and cohesive ejection becomes progressively harder. The T-Tauri Sun thus acts as a *guardian of the rheological window* via surface radiative forcing transmitted to the mantle by conduction through the silicate atmosphere.

(2) Source of stochastic perturbations. CMEs (10^{27} – 10^{29} J per event, several per day) and radiative pulsations provide a continuous source of perturbation.

(3) Obliquity perturber. CMEs acting on the proto-atmosphere and the magnetosphere deliver impulsive torques on the spin vector, contributing to the maintenance of the axial oscillation $\varepsilon \in [40^\circ, 70^\circ]$ (Section 3). The T-Tauri phase also provides an independent stopping mechanism: once the Sun reaches the main sequence (≈ 4.4 Ga), irradiation declines, τ_y increases, and the TPT window closes.

The T-Tauri Sun is therefore not the *engine* of the mechanism — gravity, the Coriolis force, the centrifugal force, Fe-Ni segregation, and the Taylor-Proudman columns are — but it is the *catalyst* that keeps the window open and sustains the axial oscillation that carries obliquity to its maxima.

7.5 Tidal forces

Solar tide. Relative amplitude $\approx 10^{-4}g_0$; contributes as a low-frequency resonant forcing on obliquity.

Growing proto-lunar tide — Mathieu resonance. Starting from Episode 2, the proto-satellite exerts a tidal force that modulates the stability of the CMT:

$$\ddot{\xi}_r = [\lambda + A_m^{(k)}(t) \cos(n_{\text{Moon}} t)] \xi_r, \quad (17)$$

which is the Mathieu equation (McLachlan, 1947). Parametric resonance bands exist for $n_{\text{Moon}} \approx 2\sqrt{|\lambda|}$ even when $\lambda < 0$: the growing proto-lunar tide is the dominant trigger of Episodes 2 through N .

Giant-planet tides. Mainly Jupiter: a low-frequency perturbation on obliquity, sustaining the oscillation over timescales of 10^5 – 10^6 years.

7.6 Parametric elliptical instability as a CMT amplifier

In a precessing fluid ellipsoid, precession flows can resonate with the fluid's inertial modes (Poincaré waves, frequencies $\omega_{\text{in}} = 2\Omega m/n$). This mechanism — the parametric elliptical instability — was established theoretically by Malkus (1968), formalized by Kerswell (2002), and demonstrated experimentally by Lacaze et al. (2004).

For the geometry of the Maclaurin spheroid at $T_{\text{rot}} = 3.5$ h and $f \approx 0.075$, the free nutation frequency is $\omega_E = \Omega f \approx 3.74 \times 10^{-5} \text{ rad s}^{-1}$ (Eq. 4). This frequency is commensurable with the fluid's inertial modes by geometric construction — not by parameter tuning. The linear growth rate is:

$$\sigma_{\text{res}} = \frac{\Omega f}{2} \approx 1.87 \times 10^{-5} \text{ rad s}^{-1}. \quad (18)$$

The CMT velocity grows exponentially:

$$U_{\text{CMT}}(t) = U_{\text{turb}} e^{\sigma_{\text{res}} t}, \quad U_{\text{turb}} \approx 0.8 \text{ km s}^{-1}, \quad (19)$$

where U_{turb} is the characteristic velocity at the Rhines scale (Rhines, 1975). The time to reach $U_{\text{crit}} \approx 9.8 \text{ km s}^{-1}$ is:

$$t_{\text{acc}} = \frac{\ln(U_{\text{crit}}/U_{\text{turb}})}{\sigma_{\text{res}}} \approx 37 \text{ h}, \quad (20)$$

with $\ln(U_{\text{crit}}/U_{\text{turb}}) = \ln(9.8/0.8) \approx 2.51$.

The CMT reaches the bifurcation velocity in ≈ 37 hours. This result is entirely determined by $f \approx 0.075$ and $\Omega = 4.99 \times 10^{-4} \text{ rad s}^{-1}$, imposed by $T_{\text{rot}} = 3.5$ h and the physics of the Maclaurin spheroid. The elliptical instability thus acts as a

powerful amplifier, bringing the CMT to the bifurcation velocity in ≈ 37 hours — a timescale fully consistent with the TPT chronology.

As Fe-Ni segregation proceeds, the density of the CMT decreases:

$$\rho_{\text{CMT}}(t) = \rho_0 - \xi_{\text{Fe}}(t)(\rho_0 - \rho_f), \quad \rho_0 \approx 4500 \text{ kg m}^{-3}, \quad \rho_f \approx 3200 \text{ kg m}^{-3}, \quad (21)$$

which amplifies all destabilizing terms via the factor $\rho_0/\rho_{\text{CMT}}(t)$ in the instability equation (Section 9.2).

7.7 Geostrophic β force and the Rhines scale

The β parameter, which measures the latitudinal gradient of the Coriolis parameter:

$$\beta = \frac{2\Omega \cos \varepsilon}{R_{\text{eq,proto}}} \approx 7.7 \times 10^{-11} \text{ m}^{-1} \text{ s}^{-1} \quad (\varepsilon = 55^\circ, R_{\text{eq,proto}} = 7.4 \times 10^6 \text{ m}), \quad (22)$$

stops the inverse cascade at the Rhines scale (Rhines, 1975):

$$L_\beta = \sqrt{\frac{U_{\text{turb}}}{\beta}} \approx 3.0 \times 10^6 \text{ m} \approx 0.47 R_\oplus, \quad (23)$$

where U_{turb} is the turbulent velocity scale. This scale $L_\beta \approx 0.47 R_\oplus$ corresponds to the harmonic mode $\ell = 1, m = 1$: the largest coherent structure possible on a sphere, and the spatial expression of the CMT.

8 Hierarchy of hypotheses

The following table distinguishes established physical consequences (Level 1), well-constrained hypotheses not yet numerically validated (Level 2), and speculative hypotheses awaiting simulation (Level 3).

Table 2: Explicit hierarchy of the hypotheses underlying the TPT.

ID	Hypothesis / Statement	Level	Depends on
H0	Central thesis: the Moon formed through three coupled transitions triggered by Fe-Ni segregation	2 (constrained)	Thermodynamic bound $E_{\text{grav}}/E_{\text{fus}} \approx 121$

ID	Hypothesis / Statement	Level	Depends on
H1	Total volumetric melting: the entire silicate mantle above the liquidus during the 4.568–4.4 Ga window	1 (established)	Factor $E_{\text{grav}}/E_{\text{fus}} \approx 121$ (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015)
H2	$\varepsilon \in [40^\circ, 70^\circ]$: sustained free nutation in the body's co-rotating frame	2 (constrained)	Absence of a Moon + 5 convergent arguments (Laskar et al., 1993a; Touma & Wisdom, 1993; Li & Batygin, 2014)
H3	The confinement of the CMT to $ \phi < 30^\circ$ admits an effective description as the fundamental mode of a nonlinear Schrödinger-type equation with potential $V_{\text{eff}}(\phi)$	3 (speculative)	Reduction from the potential vorticity equation (Redekopp, 1977; Luo, 1991; Yin et al., 2019); conditions $\tau_y \lesssim 10^2$ Pa, $\varepsilon > 45^\circ$; quantitative validity awaiting simulation (Limitation L12)
H4	The $\ell = 1$ mode is selected by the combination of BH rheology (suppression of modes $\ell < \ell_{\text{min}}$) and the inverse cascade	3 (speculative)	3D SPH simulation with full BH rheology (L1)

ID	Hypothesis / Statement	Level	Depends on
H5	$\tau_y(T)$ follows an Arrhenius law with $T^* \approx 1\,800\text{ K}$ and $\tau_y \lesssim 10^2\text{ Pa}$ for $T \gtrsim 4\,000\text{ K}$	1 (established)	Laboratory rheology (Giordano et al., 2008)
H6	The ejection threshold is defined by crossing $\lambda(U) = 0$; the generalized instability coefficient governs the transition between confinement and ejection	2 (constrained)	Coupled system (Eqs. 26, 28); P22
H7	The effective yield stress τ_y ensures the cohesive integrity of the ejected toroidal flow via the Bi_{KH} criterion	2 (constrained)	Frigaard & Nouar (2017); three independent confinement mechanisms
H8	$f_{\text{cap}} \approx 0.70$ and $\partial f_{\text{cap}} / \partial M_{\text{Moon}} > 0$	3 (speculative)	3D SPH simulation (L2); lower bound $f_{\text{cap}}^{\text{min}} \gtrsim 0.45$ (Appendix G)
H9	Dynamo delay = $350 \pm 50\text{ Ma}$, the sum of three independent phases	2 (constrained)	Hf-W (Kleine et al., 2002; Yin et al., 2002) + zircons (Wilde et al., 2001; Valley et al., 2002) + Tarduno et al. (2025)
H10	Seismic preservation window: 200–530 km	2 (constrained)	Ilmenite cumulate overturn (Briaud et al., 2023) + mare volcanism (Li et al., 2021; Che et al., 2021)

ID	Hypothesis / Statement	Level	Depends on
H11	Two-phase, 'aā'-type CMT structure: mobile plastic core + fragmented clinker crust	3 (speculative)	3D SPH simulation (L1, L3)
H12	$Bi_{KH} \gg 1$ suppresses Kelvin-Helmholtz fragmentation at large and intermediate scales	2 (constrained)	Frigaard & Nouar (2017); three independent confinement mechanisms
H13	$b' = 1$ is a dynamical attractor of the coupled segregation-ejection system	2 (constrained)	Geostrophic equilibrium + angular momentum conservation (Appendix B)
H14	The inverse cascade is halted at the Rhines scale $L_\beta \approx 0.47 R_\oplus$; the peak at $\ell \approx 14$ is redistributed toward $\ell = 1$ by nonlinearity and BH rheology	3 (speculative)	3D SPH simulation (L1)
H15	The ejected flow transits through three phases: ballistic expansion $\rightarrow$ orbital insertion $\rightarrow$ cohesive accretion beyond R_{Roche}	3 (speculative)	3D SPH simulation (L6)
H16	The parametric elliptical instability amplifies the CMT with growth rate $\sigma_{\text{res}} = \Omega f / 2$	2 (constrained)	Malkus (1968); Kerswell (2002); Lacaze et al. (2004); Maclaurin geometry
H17	The T-Tauri phase keeps $\tau_y \lesssim 10^2$ Pa via irradiation, enabling the $\ell = 1$ mode	2 (constrained)	Stefan-Boltzmann law + BH rheology; testable by simulation

9 The three coupled transitions

The single engine is the progressive segregation of Fe-Ni toward the forming core. It simultaneously governs three regime changes: the rheological transition opens the window, the mechanical transition produces the ejection episodes, and the magnetic transition closes the loop with the terrestrial dynamo.

9.1 Transition 1 — Rheological: the Bingham-Herschel law

A Newtonian fluid flows under any applied stress. A Bingham-Herschel (BH) fluid, on the other hand, resists like a solid as long as the applied stress does not exceed a plasticity threshold τ_y (Bingham, 1922; Herschel & Bulkley, 1926):

$$\tau = \tau_y + k\dot{\gamma}^n \quad (\tau > \tau_y), \quad (24)$$

where τ is the applied shear stress, τ_y the yield stress, k the consistency index, $\dot{\gamma}$ the shear rate, and n the flow index. For a partially crystallized silicate magma at a solid fraction $\phi \approx 0.1\text{--}0.3$: $n = 1.2$ (shear-thickening, Caricchi et al. 2007), $\tau_y \in [10^2, 10^4]$ Pa at $T = 3\,000\text{--}3\,500$ K (Giordano et al., 2008).

The yield stress depends exponentially on temperature (Arrhenius law):

$$\tau_y(T) = \tau_y^{(0)} \exp\left(+\frac{E_a}{RT}\right), \quad E_a \approx 150\text{--}250 \text{ kJ mol}^{-1}, \quad (25)$$

where R is the gas constant and E_a the activation energy — the positive sign in the exponential reflects the fact that τ_y *increases* as T *decreases*, as for the viscosity of a liquid (Giordano et al., 2008). At $T = 3\,000$ K: $\tau_y \approx 10^2$ Pa — the flow moves easily. At $T^* \approx 1\,800$ K, depending on the value adopted for E_a within the range $150\text{--}250 \text{ kJ mol}^{-1}$, τ_y grows to $5.5 \times 10^3\text{--}8.0 \times 10^4$ Pa — an increase of one to three orders of magnitude relative to $T = 3\,000$ K, sufficient for cohesive ejection to become progressively harder as the T-Tauri Sun exits its active phase and the surface cools.

This is the natural thermal shutdown mechanism at $T^* \approx 1\,800$ K: the TPT window closes irreversibly through the Arrhenius law alone, with no additional hypothesis. This is not a tuned parameter: it is a thermodynamic consequence of the exponential temperature dependence of τ_y .

The 'aa' structure of the CMT

The toroidal flow exhibits a two-phase architecture: a mobile, high-temperature plastic core surrounded by a fragmented clinker crust. This is not a cohesion problem: it is its guarantee. The outer crust protects the core against Kelvin-Helmholtz instabilities at the flow/atmosphere interface. A quenched-glass shell (analogous to the glassy margins of pillow lavas, Hekinian et al. 1989), forming in ≈ 7 minutes on contact with the atmosphere, completes this rheological shield. This structure is consistent with the work of Frigaard & Nouar (2017) and Balmforth & Craster (2001).

9.2 Transition 2 — Mechanical: dynamical instability and ejection

9.2.1 Principle: a dynamical instability

The ejection of matter from the CMT is a *deterministic dynamical instability*: when the net radial acceleration acting on a fluid parcel becomes repulsive, the parcel accelerates outward and leaves the torus. A parcel undergoing an infinitesimal radial displacement ξ_r obeys the Solberg-Høiland parcel-displacement equation (Solberg, 1936; Høiland, 1941):

$$\ddot{\xi}_r = \lambda \xi_r, \quad \begin{cases} \lambda < 0 & : \text{oscillation, stable confinement,} \\ \lambda > 0 & : \text{exponential growth, ejection.} \end{cases} \quad (26)$$

The ejection threshold is therefore the condition $\lambda = 0$. This threshold is a well-defined surface in the parameter space $(\Omega, \varepsilon, \alpha, \beta)$, which gives the mechanism a *predictive* character: given conditions, ejection occurs or does not occur unambiguously.

9.2.2 The stability coefficient λ

In solid rotation ($U = \Omega r$), with $U \propto r^\alpha$ the azimuthal velocity profile, the stability coefficient gathers all the radial accelerations acting on the parcel:

The instability coefficient	
$\lambda = \underbrace{-\frac{g_{\text{eff}}}{r} - N^2}_{\text{constant terms}} + \underbrace{\Omega^2 \left(\overbrace{2 \sin \varepsilon_{\text{dyn}}}^{\text{Eötvös}} - \overbrace{2(1 + \alpha)}^{\text{Rayleigh}} + \overbrace{\beta}^{\text{centrif. buoyancy}} \right)}_D \quad (27)$	

Each term has a precise physical origin — and a simple story:

- $-g_{\text{eff}}/r$: confining effective gravity, always *stabilizing*; this is the term to overcome.
- $-N^2$: *thermal* buoyancy (squared Brunt-Väisälä frequency, Väisälä, 1925; Brunt, 1927). The magma is cooled from above: the thermal gradient makes the medium superadiabatic, so $N^2 < 0$ and $-N^2 > 0$; this thermal contribution is *destabilizing* and sustains convection. The *compositional* part of the density gradient is carried separately by β (Ledoux-type separation, Appendix C).
- $2 \sin \varepsilon_{\text{dyn}}$: *non-traditional* component of the Coriolis force (Eötvös effect): any prograde flow is radially lightened by $2\Omega U \cos \phi$, where ϕ is latitude. Here ε_{dyn} is the *colatitude of the CMT relative to the instantaneous dynamical equator* ($\sin \varepsilon_{\text{dyn}} = \cos \phi$) — no orbital reference enters the internal dynamics. Neglected in the “traditional approximation” of geophysics, this component must be retained at low latitudes (Gerkema et al., 2008): this is exactly the regime of

the intertropical band. *Destabilizing* term, maximal when the torus straddles the dynamical equator ($\varepsilon_{\text{dyn}} \rightarrow 90^\circ$).

- $-2(1 + \alpha)$: Rayleigh term (Rayleigh, 1917); the epicyclic frequency $\kappa^2 = r^{-3} d[(rU)^2]/dr = 2(1 + \alpha)U^2/r^2$ is always *stabilizing*, hence the negative sign. This centrifugal-load gradient contains *by itself* the entire centrifugal effect: no separate “centrifugal pressure” term should be added to it, on pain of double counting (Appendix C).
- $+\beta$: *centrifugal buoyancy* on the density gradient, $\beta \equiv -d \ln \rho / d \ln r$. In the CMT, $\beta > 0$: Fe-Ni segregation makes the body denser at depth. With respect to gravity, this stratification is stable; but the centrifugal force is a gravity that points *outward*, and for it, the same stratification is a Rayleigh-Taylor-type configuration: the dense fluid is pushed outward. *Destabilizing* term — and essential: it is the engine’s fuel. Fe-Ni segregation thus does double duty: it accelerates rotation through contraction (skater effect) and it creates the stratification whose centrifugal buoyancy fuels the ejection.

9.2.3 The rotation threshold

Setting $\lambda = 0$ isolates the critical angular velocity:

$$\Omega_{\text{crit}}^2 = \frac{g_{\text{eff}}/r + N^2}{2 \sin \varepsilon_{\text{dyn}} - 2(1 + \alpha) + \beta}. \quad (28)$$

A finite threshold exists only if the denominator D is positive, i.e., if $\beta > 2(1 + \alpha) - 2 \sin \varepsilon_{\text{dyn}}$. The driving role of β is directly visible here: with the nominal values, $2 \sin \varepsilon_{\text{dyn}} - 2(1 + \alpha) = 1.879 - 3 = -1.12$ — without centrifugal buoyancy, no rotation would ever trigger ejection. With $\beta = 2.5$ ($\varepsilon_{\text{dyn}} = 70^\circ$ at the nutation peak, $\alpha = 0.5$, $N^2 = -1.93 \times 10^{-7} \text{ s}^{-2}$), the denominator is $D = 1.379$ and the threshold $\Omega_{\text{crit}} \approx 1.27 \Omega_0$, reachable through angular momentum input without approaching the critical rotation regime.

The sensitivity of this threshold to the rotation profile α is direct: for $\alpha \lesssim 1.0$, the threshold remains accessible via the angular momentum input of Appendix H; beyond $\alpha \approx 1.19$, the denominator D vanishes and the threshold diverges — ejection ceases to be reachable through this channel, which physically bounds the mechanism to moderately super-rigid profiles. The nominal value ($\alpha = 0.5$, $\Omega_{\text{crit}}/\Omega_0 = 1.27$) operates close to — about 1.7% above — the strict dynamical instability limit of the Maclaurin spheroid independently derived in Section 4.3 ($\Omega/\Omega_0 \approx 1.25$, corresponding to $T_{\text{rot}} \approx 2.8 \text{ h}$). This proximity may reflect a physical link between the two phenomena — the instability that triggers ejection potentially being a local manifestation of the same shape transition — rather than a fortuitous coincidence, but the derivation of λ (Appendix C) assumes an axisymmetric Maclaurin geometry whose strict validity in the immediate vicinity of this threshold remains to be confirmed by

a simulation allowing triaxial deformations.

9.2.4 The episode cycle

The mechanism is a deterministic **relaxation cycle**:

1. Ω grows slowly through net angular momentum input (skater effect from Fe-Ni segregation, supplemented by continuous accretion inputs).
2. The axis nutates (period ≈ 46.7 h): the effective colatitude ε_{dyn} oscillates within $[40^\circ, 70^\circ]$. At the nutation peak, the threshold Ω_{crit} is minimal.
3. A major Late Veneer impact (body $\gtrsim 750$ km) occurs during the post-ejection plateau window (Appendix I). The impact's impulsive torque provides a jump $\Delta\Omega$, and ε is already close to its maximum. The combination of the two lowers Ω_{crit} and crosses the threshold.
4. A tongue of magma stretches, pinches off via the Rayleigh–Plateau instability (Plateau, 1873; Rayleigh, 1878), delayed by Bingham cohesion, and detaches tangentially: an ejection episode.
5. The ejection carries away a fraction η of the angular momentum; Ω falls below the threshold, and the system then recharges until the next major impact.

9.2.5 Role of cohesion: the Bi_{KH} criterion

The Bingham yield stress τ_y **does not enter** the ejection threshold (Eq. 28): at the scale of the torus, it is negligible compared to gravity. Its role is distinct: it maintains the cohesion of the ejected flow, which propagates as a *continuous tongue*, via the Bingham–Kelvin–Helmholtz criterion $\text{Bi}_{\text{KH}} \gg 1$ (Appendix D). Flow cohesion and the ejection threshold are thus two physically separate questions.

9.2.6 Critical velocity and maximum ejectable mass

The critical ejection velocity, derived from the energetic escape condition, is:

$$U_{\text{crit}}^{(0)} = \sqrt{2 g_{\text{eff}}^{(0)} R_{\text{eq,proto}}} \approx 9.8 \text{ km s}^{-1}, \quad (29)$$

consistent with the escape-energy scale. The Mach number at ejection, using the speed of sound in the two-phase magma $c_s \approx 0.9 \text{ km s}^{-1}$ (Mader et al., 2013), is $\text{Ma} = U_{\text{crit}}/c_s \approx 10.9$: the ejection is **hypersonic**.

The maximum ejectable mass per episode is the fraction of the CMT with $U > U_{\text{crit}}$:

$$M_{\text{ej}}^{\text{max}} \approx f_{\text{inst}} M_{\text{CMT}}, \quad (30)$$

where f_{inst} is the unstable fraction of the velocity distribution. For a typical rotation

profile, this fraction is ≈ 0.1 , giving:

$$M_{\text{ej}}^{\text{max}} \approx 2.0\text{--}4.0 \times 10^{22} \text{ kg}, \quad (31)$$

consistent with the mass budget (Section 10). This is why $N = 2\text{--}3$ episodes suffice.

9.2.7 Cohesion of the ejected flow: the Bi_{KH} criterion

For a free flow, the Weber number $We \approx 1.4 \times 10^{15}$ would imply fragmentation through atomization. However, the CMT is not a free flow: it is a collective, self-gravitating toroidal flow, confined by toroidal symmetry, self-gravitation, and atmospheric pressure.

The correct criterion is the Bingham-Kelvin-Helmholtz number, which compares the yield stress to the Kelvin-Helmholtz shear stress: for $\text{Bi}_{\text{KH}} \gg 1$, the yield stress suppresses the KH instability and cohesion is maintained (Frigaard & Nouar, 2017). Three independent conditions reinforce cohesion: (1) atmospheric confinement ($P \approx 10^5\text{--}10^7$ Pa); (2) internal crystal network ($\phi \approx 0.1\text{--}0.3$, multiplying fragmentation resistance by a factor of 3 to 10); (3) density ratio $\rho_{\text{flow}}/\rho_{\text{atm}} \approx 200$ (Rayleigh-Taylor suppression). The full demonstration of the Bi_{KH} criterion is given in Appendix D.

Effective cohesion reinforcements

The effective cohesion is the product of the three reinforcements:

$$\text{Bi}_{\text{KH}}^{\text{eff}} = \text{Bi}_{\text{KH}} \times f_{\text{atm}} \times f_{\text{crist}} \times f_{\text{RT}} \quad (32)$$

with:

- $f_{\text{atm}} \sim 10\text{--}100$ (atmospheric confinement) - $f_{\text{crist}} \sim 3\text{--}10$ (internal crystal network) - $f_{\text{RT}} \sim 10\text{--}50$ (Rayleigh-Taylor suppression)

With the nominal values, $\text{Bi}_{\text{KH}} \sim 3 \times 10^{-2}$ and the product of the three reinforcements is ~ 300 , giving $\text{Bi}_{\text{KH}}^{\text{eff}} \sim 9$, much greater than 1. Cohesion is therefore assured.

Acknowledged limitation: this calculation is a nominal estimate. A 3D SPH simulation with full BH rheology is required to validate the exact values, in particular the nonlinear interaction among the three effects (Limitation L14, Section 18).

9.2.8 Thermal resistance of the ejected tongue during hypersonic flight

Since ejection occurs at $\text{Ma} \approx 10.9$ (Eq. 29), the magmatic tongue crosses the confining silicate atmosphere in a detached-shock regime, analogous — in reverse — to that of a body undergoing atmospheric re-entry. The stagnation-point shock heat flux follows the Sutton-Graves law, $q \sim k \sqrt{\rho_{\text{atm}}/R} U^3$, where R is the characteristic body radius. For the geometry of the tongue ($R \sim 10^3$ km, derived from the volume

$V = M_{\text{ej}}^{\text{max}}/\rho_{\text{flow}}$), this flux remains modest ($q \sim 10^5\text{--}10^6 \text{ W m}^{-2}$) compared to classical atmospheric re-entry heat shields, since the specific flux decreases with the square root of the nose radius.

Over the flight duration ($t_{\text{ej}} \approx 0.6 \text{ h}$), the ablated mass — estimated by integrating this flux over the entire surface of the body and comparing it to a typical ablation enthalpy for silicate ($\sim 10^7 \text{ J kg}^{-1}$) — remains below 10^{-7} of $M_{\text{ej}}^{\text{max}}$: entirely negligible relative to the mass budget (Section 10). This result is dominated by the large size of the ejected body (the surface-to-volume ratio, and hence ablation efficiency, decreasing with R) rather than by an active protection mechanism: much as large bolides survive atmospheric re-entry proportionally better than small meteors, the ejected tongue, of planetary dimension, ablates negligibly over its flight time.

The quenched glassy crust (Section 9.1), already formed before ejection to ensure flow cohesion against Kelvin-Helmholtz instability, nonetheless constitutes an additional safeguard against local thermal-flux heterogeneities — at the leading edge, or within wake turbulence — not captured by this average shock-flux estimate.

9.2.9 Transient post-ejection precession and crustal dichotomy

The ejection of $M_{\text{ej}} \approx 2\text{--}4 \times 10^{22} \text{ kg}$ from the intertropical band is not dynamically negligible. The asymmetric mass loss modifies the inertia tensor, producing: (1) a decrease $\Delta\Omega/\Omega \approx -10\text{--}15\%$, lowering g_{eff} and U_{crit} for the next episode; (2) shape oscillations as the spheroid contracts toward a rounder sphere; (3) a reorganization of obliquity toward a new value within $[40^\circ, 70^\circ]$.

This last point has a direct geometric, not merely numerical, consequence: since the intertropical band $|\phi| < 30^\circ$ is defined perpendicular to the instantaneous axis $\Omega(t)$ (Section 3), a new value of ε shifts the position of the dynamical equator $\phi = 0$ itself, and hence the entire active band, relative to the body. The intertropical band hosting the second episode is therefore not, in general, the same region of the body that hosted the first: same angular width ($\pm 30^\circ$), same definition relative to $\Omega(t)$, but a different location on the spheroid. It is this episode-by-episode geometric displacement of the active band that constitutes the direct cause of the crustal asymmetry described below, and not merely the change in the numerical value of ε .

This displacement of the active band, combined with the shape oscillations of point (2) above, has a further consequence for the CMT itself: the torus that reforms for the next episode does not reproduce an identical copy of the previous one simply relocated. It forms on a spheroid whose own shape has slightly changed, along a band whose position has itself changed. The next CMT is therefore ejected toroidally in a manner distinct from the previous one, both in shape and orientation. This remains, at this stage, a qualitative statement: no quantitative derivation of the shape or angular offset between successive episodes is attempted here.

The second episode ejects a cohesive toroidal flow onto a still-hot, deformable proto-

POB, within a limited solid angle (corresponding to the projected intertropical band), which becomes the present far side. Three mechanisms amplify the resulting crustal asymmetry: early gravitational locking ($\tau_{\text{lock}} \sim 10^4\text{--}10^5$ years (Wieczorek et al., 2013)), the residual obliquity of the proto-POB, and irreversible BH plastic welding. The thickness ratio $\approx 2:1$ (near side ≈ 34 km versus far side ≈ 54 km, Wieczorek et al. 2013) is a qualitative prediction of the TPT.

9.3 Transition 3 — Magnetic: the progressive dynamo

The ≈ 350 Ma delay between the inferred time of lunar formation (≈ 4.55 Ga) and the first geological detection of Earth’s magnetic field (≈ 4.2 Ga, Tarduno et al. 2025) is not a tuned parameter in the TPT. It is proposed as the sum of three independently constrained durations:

$$\Delta t_{\text{dynamo}} = \underbrace{\Delta t_{\text{ej}}}_{\approx 60 \text{ Ma}} + \underbrace{\Delta t_{\text{refr}}}_{\approx 140 \text{ Ma}} + \underbrace{\Delta t_{\text{em}}}_{\approx 150 \text{ Ma}} \approx 350 \text{ Ma.} \quad (33)$$

Phase 1 (≈ 60 Ma) — Active Fe-Ni segregation. Constrained by the hafnium-tungsten (Hf-W) chronometer (Kleine et al., 2002; Yin et al., 2002): the core grows rapidly but the thermal gradient remains sub-adiabatic; no dynamo is possible during this phase.

This ≈ 60 Ma timescale concerns the *global* segregation of the planetary core — the cumulative build-up of metallic iron at the center of the entire body — and not the replenishment of each ejection episode, which operates on a far more restricted spatial and temporal scale. The local density stratification within the active band, the one that feeds the β term in Eq. 27, is renewed by Stokes sedimentation ($\tau_{\text{Stokes}} \approx 254$ s, Appendix F) and by convective stirring of the CMT ($U_{\text{turb}} \approx 0.8 \text{ km s}^{-1}$, Eq. 19) over a characteristic distance far exceeding the thickness of the active band during the recharge window ($\approx 10\text{--}12$ days). The local replenishment required by each ejection cycle is therefore not limited by the slow global segregation of the core: these are two distinct clocks, nested within one another.

Phase 2 (≈ 140 Ma) — Protocrust formation. The growing Moon stabilizes obliquity at $\approx 23.5^\circ$; the protocrust forms around 4.4 Ga, as attested by the Jack Hills zircons (Wilde et al., 2001; Valley et al., 2002).

Phase 3 (≈ 150 Ma) — Progressive dynamo emergence. Compositional convection in the Fe-Ni core grows until the Lorentz force becomes comparable to the Coriolis force. The Elsasser number Λ balances these two forces. The threshold $B_{\text{crit}} \approx 2.6$ mT is derived in Appendix E. Modern geodynamo simulations (Christensen & Aubert, 2006; Olson & Christensen, 2006) show that the transition is progressive, not abrupt: $\Lambda \sim 1$ is an order-of-magnitude reference, not a strict threshold.

Section summary

The ≈ 350 Ma delay involves no free parameter: Δt_{ej} is constrained by the Hf-W chronometer (Kleine et al., 2002; Yin et al., 2002); Δt_{refr} is constrained by the Jack Hills zircons at 4.4 Ga (Wilde et al., 2001; Valley et al., 2002); and Δt_{em} is constrained by Tarduno et al. (2025) at 4.2 Ga. These are three independent observations, not an interpolation. The associated uncertainty is ± 50 Ma (Limitation L4, Section 18).

10 Mass budget: $N = 2\text{--}3$ episodes

10.1 Corrected target lunar mass

A fundamental point: the currently observed lunar mass $M_{\text{Moon}} = 7.34 \times 10^{22}$ kg is *not* the mass the TPT mechanism must produce. The Moon accreted additional mass after the ejection episodes: a differentiated body about 260 km in diameter struck the Moon at the South Pole–Aitken basin and *stayed there*, contributing $M_{\text{SPA}} \approx 3.2 \times 10^{19}$ kg (Wakita et al., 2026) — this matter is now part of the observed lunar mass — and a documented late accretion contributed an additional mass ΔM_{late} (Zellner, 2019). The target mass for the TPT mechanism is therefore:

$$M_{\text{TPT}} = M_{\text{Moon}} - M_{\text{SPA}} - \Delta M_{\text{late}} < 7.34 \times 10^{22} \text{ kg.} \quad (34)$$

10.2 Mass budget equation

The mass budget equation

$$M_{\text{Moon}} = N \cdot f_{\text{cap}} \cdot M_{\text{ej}}^{\text{max}} + M_{\text{SPA}} + \Delta M_{\text{late}}, \quad (35)$$

with $N = 2\text{--}3$ (nominal), $f_{\text{cap}} \in [0.65, 0.85]$ (nominal 0.70), $M_{\text{ej}}^{\text{max}} \approx 2.0\text{--}4.0 \times 10^{22}$ kg per episode.

f_{cap}	$M_{\text{TPT}}^{(N=2)}$	$M_{\text{TPT}}^{(N=3)}$	Fraction of M_{Moon}
0.50	2.5×10^{22} kg	3.7×10^{22} kg	34–50%
0.70	3.5×10^{22} kg	5.2×10^{22} kg	48–71%
0.85	4.3×10^{22} kg	6.4×10^{22} kg	59–87%

The remainder is accounted for by M_{SPA} and documented late accretion. The budget is robust across the entire f_{cap} range.

f_{cap} is a growing state variable: $\partial f_{\text{cap}} / \partial M_{\text{Moon}} > 0$, since the larger the proto-Moon, the more efficiently it captures subsequent ejecta. This positive feedback, combined with the decrease of $g_{\text{eff}}^{(k)}$ at each episode, defines a dynamical attractor that reinforces the mechanism.

Feedback during TPT episodes. This growth of f_{cap} with the mass already accreted can be formalized, to first order, by a linear law:

$$f_{\text{cap}}(M_{\text{proto}}) = f_0 + \alpha_f M_{\text{proto}}, \quad \alpha_f > 0, \quad (36)$$

where M_{proto} is the proto-Moon mass already in place before the episode considered. In steady state (after the last episode, $M_{\text{proto}} \simeq M_{\text{Moon}} - \Delta M_{\text{late}}$), the mass budget (35) closes into an implicit equation for ΔM_{late} , whose sensitivity to the nominal value f_0 is $\partial \Delta M_{\text{late}} / \partial f_0 = -NM_{\text{ej}}^{\text{max}} / (1 - NM_{\text{ej}}^{\text{max}} \alpha_f)$. With $NM_{\text{ej}}^{\text{max}} \sim 7.5 \times 10^{22}$ kg, a 0.1 variation in f_0 changes ΔM_{late} by about 7.5×10^{21} kg, i.e., $\sim 10\%$ of the lunar mass. This scaling law does not predict α_f *ab initio* — its value is a matter for 3D SPH simulation (Limitation L2) — but it formalizes the feedback already qualitatively established above and physically bounds α_f through the condition $NM_{\text{ej}}^{\text{max}} \alpha_f < 1$, required for the budget to remain well-posed.

This feedback effect *during* the episodes is distinct from, and cumulative with, the geometric scaling law of late accretion *after* the ejections end ($\Delta M_{\text{late}} \propto M_{\text{TPT}}^{2/3}$, Section 10.3): the former acts in real time on the capture of the mechanism's own successive ejecta, the latter on the subsequent capture of external matter by the already-formed body. The final budget is the sum of these two contributions.

A sensitivity analysis varying f_{cap} and $M_{\text{ej}}^{\text{max}}$ simultaneously over their respective ranges (Section 10) confirms $N = 2\text{--}3$ for the nominal case as well as for any favorable combination of the two parameters. The most unfavorable corner of this range — f_{cap} at its lower bound combined with $M_{\text{ej}}^{\text{max}}$ at its lower bound — formally pushes N beyond this range, with a correspondingly larger mass deficit ΔM_{late} to be covered by late accretion.

This larger deficit is not, however, freely compensable, since the capture capacity of late accretion itself depends on the mass of the capturing body: the geometric cross section grows as $M_{\text{TPT}}^{2/3}$, so that $\Delta M_{\text{late}} \propto M_{\text{TPT}}^{2/3}$ at fixed impact velocity and external mass flux. Applying this scaling law between the nominal case ($f_{\text{cap}} = 0.70$, $N = 3$, $M_{\text{TPT}} \approx 5.2 \times 10^{22}$ kg, $\Delta M_{\text{late}} \approx 2.13 \times 10^{22}$ kg) and the most unfavorable corner of the official range ($f_{\text{cap}} = 0.65$, $M_{\text{ej}}^{\text{max}} = 2.0 \times 10^{22}$ kg, $N = 3$, $M_{\text{TPT}} \approx 3.9 \times 10^{22}$ kg, requiring $\Delta M_{\text{late}} \approx 3.45 \times 10^{22}$ kg), the late accretion mass actually available in this unfavorable corner reduces to $\Delta M_{\text{late}} \approx 1.76 \times 10^{22}$ kg — leaving a residual deficit of about 1.7×10^{22} kg ($\approx 23\%$ of M_{Moon}) uncovered. This unfavorable parameter combination is thus doubly penalized by this mechanism, rather than compensated by a simple adjustment of late accretion: a low capture efficiency simultaneously reduces the direct TPT contribution and the body's subsequent capacity to accrete the missing mass. $N = 2\text{--}3$ remains the nominal and most parsimonious result of the model; the most unfavorable corner of the parameter range remains, under this scaling law, physically insufficient to reach the observed lunar mass without an as-yet undocumented additional mass source.

10.3 f_{cap} as a testable prediction, not a free parameter

The position adopted here is that f_{cap} should not be treated as an adjustable parameter whose wide range would make the mass budget uninformative, but as a quantity that the TPT predicts quantitatively and which can, in principle, falsify the mechanism.

Physical basis: the classical disk-accretion interval. The reference N-body simulations of lunar accretion from an impact-generated debris disk (Kokubo et al., 2000) establish that the efficiency of incorporating disk material into a final moon is 10 to 55%, this efficiency increasing with the initial specific angular momentum of the disk. This result, robust across a wide range of initial conditions and numerical resolutions, constitutes the external reference against which f_{cap} must be compared.

The gap as a potential signature. The nominal value $f_{\text{cap}} \approx 0.70$ adopted in the TPT is markedly above the upper bound of this classical interval (55%). Within the TPT framework, this gap is not accidental: it is attributed to the coherence of the ejected toroidal flow and its orbital concentration within a limited solid angle (Appendix G), unlike a debris disk dispersed over 4π steradians as in the classical giant-impact scenario. If this interpretation is correct, the gap between the f_{cap} value adopted by the TPT and the classical interval becomes a discriminating test rather than a mere parametric latitude.

Falsifiable prediction — P17 — Capture efficiency beyond the classical disk-accretion interval

The orbital coherence of the toroidal flow ejected by the TPT pushes the effective capture efficiency beyond the classical interval of 10–55% established for accretion from a dispersed debris disk (Kokubo et al., 2000). The TPT predicts $f_{\text{cap}} \gtrsim 0.65\text{--}0.70$.

Falsification: if a value of f_{cap} derived from a dedicated 3D SPH simulation of the CMT geometry (BH rheology, atmospheric confinement, orbital concentration within the intertropical band) falls within the classical interval of 10–55% rather than beyond it, this aspect of the theory is falsified. An f_{cap} measured within the classical interval would not necessarily invalidate the TPT as a whole (the mass budget would remain achievable via M_{SPA} and late accretion), but it would strip f_{cap} of its status as a distinctive signature relative to the giant-impact model.

Acknowledged limitation: this prediction rests on the hypothesis that the orbital concentration of the TPT flow (Appendix G) produces an accretion regime qualitatively different from the classical dispersed disk. This hypothesis is itself Level 3 (not quantitatively demonstrated), so this only becomes a robust test once the 3D SPH simulation is available. In the meantime, this remains a qualified prediction: the gap with the classical interval is consistent with the TPT, but does not independently confirm it.

11 Angular momentum transport by the ejecta

In the present framework, the loss of angular momentum is not treated as a single-velocity ballistic process, but as a distributed transport mechanism resulting from a spectrum of ejected matter generated over successive phases driven by differentiation.

11.1 Ejecta distribution function

We describe the mass distribution of the ejecta as a function of velocity and emission angle:

$$dM = \rho(v, \theta) dv d\theta, \quad (37)$$

where v is the ejection velocity, θ is the angle relative to the local tangential direction, and $\rho(v, \theta)$ is the phase-dependent ejecta distribution function.

The distribution $\rho(v, \theta)$ is not arbitrary. It follows from three physical processes: the CMT velocity spectrum is not monokinetic, turbulent fluctuations at the Rhines scale (Section 7.6) producing a velocity spread $\delta v_{\text{CMT}} \sim U_{\text{turb}} \approx 0.8 \text{ km s}^{-1}$; and the finite extent of the torus produces an angular spread $\rho_\theta(\theta) \propto \cos \theta$ for $|\theta| < \theta_{\text{max}}$, with $\theta_{\text{max}} \approx 30^\circ$.

For analytical tractability, we adopt the separable form:

$$\rho(v, \theta) = M_{\text{ej}} \mathcal{N}_v \mathcal{N}_\theta \exp \left[-\frac{(v - \bar{v})^2}{2\sigma_v^2} \right] \cos \theta, \quad v > v_{\text{esc}}, |\theta| < \theta_{\text{max}}, \quad (38)$$

where $v_{\text{esc}} = \sqrt{2g_{\text{eff}}R_p}$. No free parameter is introduced beyond those already present in the theory developed in the previous sections.

11.2 Specific angular momentum

The specific angular momentum of an ejecta element is $h = r v \sin \theta$, where r is the effective emission radius. This radius is not simply the planetary radius R_p : the silicate atmosphere (Section 2) extends the effective surface to $r_{\text{atm}} \approx R_p + H$, where the scale height $H \approx 200 \text{ km}$ for $T \approx 3000 \text{ K}$ and $g \approx 6 \text{ m/s}^2$; and the toroidal flow is raised above the surface by centrifugal effects, with the center of the CMT located at $r_{\text{CMT}} \approx R_p + U_{\text{CMT}}^2 / (2g_{\text{eff}}) \approx R_p + 500 \text{ km}$. We therefore adopt $r \approx R_p + \Delta r$, with $\Delta r \approx 300\text{--}500 \text{ km}$. This correction introduces only a few percent of uncertainty in the angular momentum budget, well within the overall margin of error.

The mass of the silicate atmosphere itself ($M_{\text{atm}} \lesssim 10^{21} \text{ kg}$, obtained from $M_{\text{atm}} \sim 4\pi R_p^2 P / g$ for $P \sim 10^5\text{--}10^7 \text{ Pa}$) is negligible in both budgets: it represents less than 2% of the target lunar mass in the mass budget (Section 10), and its contribution to the moment of inertia ($I_{\text{atm}} \sim \frac{2}{3} M_{\text{atm}} R_p^2$) and to the initial angular momentum L_{init} remains, in both cases, below 0.1% of the corresponding quantities.

11.2.1 Assembly beyond the Roche radius

An apparent paradox deserves resolution here, as it conditions the very birth of the Moon. The maximum specific angular momentum an ejecta can carry at launch, $h_{\max} \approx 7.74 \times 10^{10} \text{ m}^2 \text{ s}^{-1}$, corresponds to a circular orbit of radius $r_{\text{circ}} = h_{\max}^2 / (GM_p) \approx 1.5 \times 10^7 \text{ m}$, i.e., about 15,000 km — *below* the Roche radius ($\approx 22,200 \text{ km}$), where tidal forces forbid the direct accretion of a self-gravitating body (Roche, 1849). Taken in isolation, no captured ejecta could therefore become a moon: the matter seems doomed to orbit as a ring, torn apart by tides.

The resolution of this paradox is known, and it requires no new hypothesis: the captured torus is not a set of particles frozen on independent Keplerian orbits, but a dense disk where inelastic collisions and gravitational interactions produce an effective viscosity. This viscosity does what any disk viscosity does: it transports angular momentum outward and mass in both directions, so that a fraction of the disk is pushed beyond the Roche radius — where accretion becomes possible again and where the proto-Moon assembles, fed by the edge of the disk. This spreading mechanism is precisely the one that builds the Moon in post-impact disk models: in the particulate regime, assembly is completed within a few months to a year (Ida et al., 1997); in the fluid-disk regime, where radiative cooling regulates viscosity, it extends over $\sim 10^2$ – 10^3 years (Salmon & Canup, 2012). In both cases, assembly is completed well before the encounter with the evection resonance (10^3 – 10^5 years, Section 11.5): the Moon exists as a single body by the time the angular momentum sink engages.

This result has an important conceptual consequence for the TPT: the cohesion of the ejected flow (the Bi_{KH} number, Appendix D, Limitation L14) governs the *launch and transport* phase — how much matter reaches orbit, and with what angular momentum — but it is not required for the *assembly* phase, which viscous spreading accomplishes on its own. The critical question posed to the SPH simulation is therefore not “does the flow remain perfectly cohesive all the way to orbit?” but “what fraction of the ejected mass is captured into stable orbit?” — that is, the determination of f_{cap} (Limitation L2).

11.2.2 Energy budget of the ejection

With the angular momentum budget now established, the symmetric question arises naturally: does the proto-Earth have the *energy* needed to launch three magmatic tongues at escape velocity? The total energy bill of the ejecta is:

$$E_{\text{kin,ej}} = \frac{1}{2} N M_{\text{ej}} \bar{v}^2 \approx \frac{1}{2} \times 3 \times 2.5 \times 10^{22} \times (9.8 \times 10^3)^2 \approx 3.6 \times 10^{30} \text{ J}, \quad (39)$$

most of which is spent climbing the potential well ($\bar{v}$ being close to escape velocity, the residual kinetic energy at infinity is small). The available reservoir is the

rotational energy of the proto-Earth:

$$E_{\text{rot}} = \frac{1}{2} I_p \Omega_p^2 \approx \frac{1}{2} \times 1.32 \times 10^{38} \times (4.99 \times 10^{-4})^2 \approx 1.6 \times 10^{31} \text{ J}. \quad (40)$$

The ratio $E_{\text{kin,ej}}/E_{\text{rot}} \approx 0.22$: the three episodes consume about a fifth of the spin reservoir.

The total angular momentum carried away *at launch* by all the ejecta, whether subsequently captured or escaped, is:

$$L_{\text{ej}} = M_{\text{ej}} r \bar{v} \langle \sin \theta \rangle \approx 7.5 \times 10^{22} \times 7.9 \times 10^6 \times 9.8 \times 10^3 \times 0.25 \approx 1.5 \times 10^{33} \text{ J} \cdot \text{s}, \quad (41)$$

where $\langle \sin \theta \rangle \approx 0.25$ is the mean geometric factor accounting for the torus's angular spread ($\theta_{\text{max}} \approx 30^\circ$). This quantity L_{ej} is distinct from L_{loss} as formally defined in Eq. 44 (Section 11.3): L_{ej} covers the whole ejecta at the moment of launch, whereas L_{loss} covers only the uncaptured fraction. Weighting by the escaped fraction $(1 - f_{\text{cap}}) = 0.30$, one obtains a mean estimate of L_{loss} consistent with this formal definition:

$$L_{\text{loss}} \approx (1 - f_{\text{cap}}) L_{\text{ej}} \approx 0.30 \times 1.5 \times 10^{33} \approx 4.5 \times 10^{32} \text{ J} \cdot \text{s}. \quad (42)$$

This value is numerically below the conservative bound $L_{\text{esc}}^{\text{max}} \approx 1.74 \times 10^{33} \text{ J} \cdot \text{s}$ (Section 11.5), which instead assumes purely tangential ejection ($\sin \theta = 1$) for the escaped matter alone — a deliberately generous upper bound, not a central value. The escape channel therefore covers an even smaller fraction of the deficit ΔL than the conservative bound alone would suggest.

The rotational energy loss associated with L_{ej} is:

$$\Delta E_{\text{rot}} \approx 2 \frac{L_{\text{ej}}}{L_{\text{init}}} E_{\text{rot}} \approx 2 \times \frac{1.5 \times 10^{33}}{6.59 \times 10^{34}} \times 1.6 \times 10^{31} \approx 7.3 \times 10^{29} \text{ J}, \quad (43)$$

to which is added, and which it is far outweighed by, the Fe-Ni segregation heat ($\approx 3 \times 10^{30} \text{ J}$, Section 9.2), which constitutes the main source covering the energy bill of ejection and otherwise sustains the molten state and convection. The energy budget is therefore covered, to order unity, by reservoirs already accounted for in the model: ejection is energetically feasible without a hidden source, and without compromising the rotational stability of the host body.

11.3 Loss of angular momentum

The total angular momentum carried away by unaccreted matter is:

$$L_{\text{loss}} = \int \int (1 - f_{\text{cap}}(v, \theta)) r v \sin \theta \rho(v, \theta) dv d\theta. \quad (44)$$

The capture efficiency $f_{\text{cap}}(v, \theta)$ is not a constant: it depends on the orbital fate of each ejecta element,

$$f_{\text{cap}}(v, \theta) = \frac{1}{2} \left[1 + \text{erf} \left(\frac{v \sin \theta - v_{\text{crit}}}{\sigma_f} \right) \right], \quad (45)$$

where $v_{\text{crit}} \approx v_{\text{circ}} = \sqrt{g_{\text{eff}} R_p}$ is the circular orbital velocity and σ_f a smoothing width representing the finite transition between captured and lost matter. Rather than treating f_{cap} as a single ad hoc value, we express it here as a function $f_{\text{cap}}(v, \theta)$ of orbital mechanics. The nominal value $f_{\text{cap}} \approx 0.70$, independently estimated from the cohesive fraction of the CMT and the re-accretion of co-orbital fragments, is consistent with the mass-weighted average of this function over the proposed ejecta spectrum.

11.4 Discrete formulation and Monte Carlo implementation

For a numerical implementation, the expression can be discretized as:

$$L_{\text{loss}} = \sum_i m_i r_i v_i \sin \theta_i (1 - f_{\text{cap},i}). \quad (46)$$

A stochastic treatment can implement this discrete formulation via Monte Carlo sampling of $\rho(v, \theta)$: sample N particles from $\rho(v, \theta)$, assign each particle a mass $m_i = M_{\text{ej}}/N$, compute $h_i = r_i v_i \sin \theta_i$, compute $f_{\text{cap},i}$ from Eq. 45, and sum $L_{\text{loss}} = \sum_i m_i h_i (1 - f_{\text{cap},i})$. This Monte Carlo approach is recommended for the 3D SPH validation (Limitation L2): it provides a robust check on the analytical estimates.

In a multi-phase formation scenario, the ejecta are further grouped into k distinct formation phases:

$$L_{\text{loss}} = \sum_k \sum_i m_{k,i} r_{k,i} v_{k,i} \sin \theta_{k,i} (1 - f_{\text{cap},k,i}), \quad (47)$$

each phase k having its own mean ejection velocity $\bar{v}_k$ (decreasing as the proto-Earth loses mass and angular momentum), its own angular spread $\theta_{\text{max},k}$ (which may increase as the CMT reorganizes between episodes), and its own mean capture efficiency $f_{\text{cap},k}$ (which increases as the proto-Moon grows, the positive feedback of Section 10.3).

11.5 Closure of the angular momentum budget

The total angular momentum budget reads:

$$L_{\text{init}} = L_{\text{TL}} + L_{\text{esc}} + L_{\text{tidal}} + L_{\text{other}}, \quad (48)$$

where:

- $L_{\text{init}} = 6.59 \times 10^{34}$ J s is the initial angular momentum of the proto-Earth at $T = 3.5$ h;
- $L_{\text{TL}} = 3.47 \times 10^{34}$ J s is the current angular momentum of the Earth–Moon system (Earth spin + lunar orbit);
- L_{esc} is the angular momentum carried away by ejecta that permanently escape the system;
- L_{tidal} is the angular momentum removed via tidal recession and the evection resonance;
- L_{other} groups losses through other channels (magnetic torque, solar wind, interaction with the protoplanetary disk).

Conservative bound on L_{esc} . For ejecta launched from $r \approx 7.9 \times 10^6$ m at $v \approx 9.8$ km/s (pre-shock velocity), the maximum specific angular momentum is $h_{\text{max}} = r\bar{v} = 7.74 \times 10^{10}$ m²/s. With a total ejected mass $M_{\text{ej}} = 7.5 \times 10^{22}$ kg and a capture efficiency $f_{\text{cap}} = 0.70$, the escaped mass is $M_{\text{esc}} = 2.25 \times 10^{22}$ kg. The upper bound on L_{esc} is therefore:

$$L_{\text{esc}}^{\text{max}} = M_{\text{esc}} h_{\text{max}} \approx 1.74 \times 10^{33} \text{ J} \cdot \text{s}. \quad (49)$$

This bound is computed using the pre-shock velocity. Since the ejection is hypersonic ($\text{Ma} \approx 11$, Section 9.2), a detached shock forms at the CMT interface, reducing the tangential velocity of the ejecta. The actual value of L_{esc} is therefore probably lower than this bound. It should be understood as a conservative estimate — an upper limit, not a central value.

Deficit and the role of the evection resonance. The deficit relative to the current state is:

$$\Delta L = L_{\text{init}} - L_{\text{TL}} - L_{\text{esc}}^{\text{max}} \approx 6.59 \times 10^{34} - 3.47 \times 10^{34} - 1.74 \times 10^{33} \approx 2.95 \times 10^{34} \text{ J} \cdot \text{s}, \quad (50)$$

or about 44.8% of the initial angular momentum.

With $L_{\text{esc}} \leq 1.74 \times 10^{33}$ J s, the escape channel covers only a minor fraction of the deficit. Most of ΔL must therefore be removed by an external sink.

The evection resonance (Ćuk & Stewart, 2012; Ćuk et al., 2016) is the most studied quantified candidate in the literature. It transfers the excess angular momentum of the Earth–Moon system to the heliocentric orbit during tidal recession. Its efficiency depends on the overall quality factor Q_{global} of the proto-Earth, which is controlled by dissipation throughout the whole body, not by the local rheology of the CMT.

11.5.1 Distinction between local and global quality factors

The evection resonance (Ćuk & Stewart, 2012; Ćuk et al., 2016) is the most studied quantified mechanism for removing this deficit. Its efficiency depends on the overall quality factor Q_{global} of the proto-Earth, which governs tidal dissipation integrated over the entire body.

For a rotation period of about 3.5 h, the characteristic tidal frequency is $\omega \simeq 5 \times 10^{-4} \text{ s}^{-1}$. In a Maxwell viscoelastic model, maximum dissipation is obtained when $\omega\tau \simeq 1$, i.e., $\tau \simeq 2 \times 10^3 \text{ s}$. For an effective shear modulus of a few GPa, this corresponds to a viscosity of about $10^{13} \text{ Pa}\cdot\text{s}$. The viscosities expected for a magma ocean, even partially crystallized ($\phi \sim 0.1\text{--}0.3$), are several orders of magnitude below this value. These regions are therefore far from the viscoelastic dissipation maximum and should not be the main contributors to the global tidal dissipation.

This point can be made quantitative. Using the Arrhenius-derived *molecular* viscosity of the silicate magma itself ($\eta \sim 10^6\text{--}10^8 \text{ Pa}\cdot\text{s}$ over the relevant cooling range, Eq. 25) in the Maxwell relation $Q^{-1} = \omega\eta/\mu$ gives a tidal timescale $\tau_{\text{tide}} \sim \mu/(\omega^2\eta) \sim 10\text{--}10^3 \text{ yr}$ — far too short to sustain the resonance over the $10^3\text{--}10^5 \text{ yr}$ required for capture (Ćuk & Stewart, 2012; Tian et al., 2017), and inconsistent with the multi-Ma cooling window itself. The molecular viscosity of the magma, whatever its exact value within the range imposed by cooling, therefore cannot be the operative dissipation channel: it is either too fluid (far below the Maxwell peak, as for the CMT’s local rheology, Section 11.5 above) or, even at its most viscous Hadean values, still well short of the $\sim 10^{13} \text{ Pa}\cdot\text{s}$ required to approach that peak. This reinforces, rather than contradicts, the turbulent-convection argument developed above: it is the effective viscosity of large-scale convective stirring ($\eta_{\text{turb}} \sim 10^{13} \text{ Pa}\cdot\text{s}$, from U_{turb} and L_{β}), and not the molecular viscosity of the quiescent magma at any temperature, that can plausibly approach the Maxwell dissipation peak and provide a Q_{global} compatible with the resonance capture timescale.

The TPT is thus compatible with a Q_{global} high enough to allow the evection resonance to act. However, the quantitative determination of Q_{global} requires dedicated thermo-rheological modeling, coupling the thermal evolution of the magma ocean to the viscoelastic response of the mantle. This modeling constitutes a testable prediction of the model (Limitation L15).

11.5.2 A common solar clock: T-Tauri mass loss and the evection resonance encounter window

The synchronization between the favorable dissipation window and the encounter with the evection resonance is not necessarily accidental: it could be set by a single common clock, the evolution of the T-Tauri Sun itself.

Stellar mass loss by wind during the T-Tauri phase is documented at a rate of order $\dot{M}_{\text{wind}} \sim 0.1 \dot{M}_{\text{accr}}$, where $\dot{M}_{\text{accr}} \sim 10^{-8} M_{\odot} \text{ yr}^{-1}$ is the typical accretion rate observed for classical T-Tauri stars (Hartmann et al., 1998; Gullbring et al., 1998), i.e., $\dot{M}_{\text{wind}} \sim 10^{-9} M_{\odot} \text{ yr}^{-1}$ (Königl & Ruden, 1993; Cranmer, 2009). Integrated over the active lifetime of the protoplanetary disk (1–10 Myr, Haisch et al., 2001), this rate gives a fractional mass loss

$$\frac{\Delta M}{M_{\odot}} \sim 10^{-3} - 10^{-2}, \quad (51)$$

consistent with the theoretical virial upper bound on total pre-main-sequence mass loss (a few tenths of $M_{\odot}$, Weidenschilling, 1978).

By conservation of orbital angular momentum during slow mass loss from the central star ($a M_{\star} \approx \text{const}$), this mass loss adiabatically shifts the Earth’s semi-major axis, and therefore the length of the sidereal year that defines the evection resonance condition (precession of the lunar perigee equal to one year):

$$\frac{\Delta T_{\text{year}}}{T_{\text{year}}} \sim \frac{\Delta M}{M_{\odot}} \sim 10^{-3} - 10^{-2}, \quad (52)$$

obtained by combining Kepler’s third law ($T^2 \propto a^3 / M_{\star}$) with $\Delta a / a \approx \Delta M / M_{\odot}$.

This drift remains slow compared to the capture/crossing time of the resonance itself ($10^3 - 10^5$ yr, Ćuk & Stewart, 2012; Tian et al., 2017): it therefore does not disturb an ongoing capture, but it determines *when*, within the 30–50 Ma active window, the resonance condition is encountered. Simultaneously, the declining luminosity of the same Sun as it exits its T-Tauri phase drives the cooling and progressive stiffening of the proto-Earth, a thermal-orbital coupling already initiated by Rufu & Canup (2020) in their variable- Q tidal evolution model. If the encounter with the resonance — $10^3 - 10^5$ yr after lunar formation — occurs early within the 30–50 Ma window, while a favorable dissipative regime is still active, closure of the angular momentum budget becomes more plausible than a static examination would suggest. This synchronization remains qualitative as it stands: it rests on the comparison of compatible timescales, not on a full coupled dynamical calculation, and constitutes a proposed extension of Limitation L15.

Budget status. In the manuscript’s current state, the angular momentum budget is not closed for the nominal value of Q_{CMT} associated with the partially molten band ($Q_{\text{CMT}} \sim 10^4$). However, this implies no contradiction, since Q_{global} is a distinct quantity that can be much larger. The tension identified in earlier versions

of this work is thus resolved by the distinction between local and global quality factors. The budget will be definitively closed once a dedicated thermo-rheological simulation provides a robust value of Q_{global} for the Hadean proto-Earth. This is the highest-priority numerical validation task (Limitation L15, Section 18).

The minimum period compatible with the escape bound, given for reference, is:

$$T_{\text{init}} \gtrsim \frac{2\pi I_p}{L_{\text{TL}} + L_{\text{esc}}^{\text{max}}} \approx \frac{2\pi(1.32 \times 10^{38})}{3.47 \times 10^{34} + 1.74 \times 10^{33}} \approx 6.33 \text{ h.} \quad (53)$$

This path is excluded by the mechanism itself: the ejection threshold Ω_{crit} (Section 9.2) is an absolute quantity, and a proto-Earth at $T = 6.33 \text{ h}$ could never reach it.

11.6 Scaling laws and dimensionless form

The angular momentum transport problem admits a natural non-dimensionalization that reveals the essential physical dependencies and simplifies the parameter space. We define the characteristic velocity scale $U_0 = U_{\text{crit}} \approx 9.8 \text{ km s}^{-1}$, the length scale $R_0 = R_p \approx 7.4 \times 10^6 \text{ m}$, the mass scale $M_0 = M_{\text{ej}} \approx 2.5 \times 10^{22} \text{ kg}$, and the angular momentum scale $H_0 = M_0 R_0 U_0 \approx 1.8 \times 10^{36} \text{ kg m}^2 \text{ s}^{-1}$. The dimensionless variables $\tilde{v} = v/U_0$, $\tilde{r} = r/R_0$, $\tilde{m} = m/M_0$, and $\tilde{h} = h/(R_0 U_0) = \tilde{r}\tilde{v} \sin \theta$ recast the ejecta distribution as:

$$\tilde{\rho}(\tilde{v}, \theta) = \tilde{M}_{\text{ej}} \tilde{\mathcal{N}}_v \tilde{\mathcal{N}}_\theta \exp \left[-\frac{(\tilde{v} - \tilde{v})^2}{2\tilde{\sigma}_v^2} \right] \cos \theta, \quad \tilde{v} > \tilde{v}_{\text{esc}}, \quad |\theta| < \theta_{\text{max}}, \quad (54)$$

with $\tilde{M}_{\text{ej}} = 1$ by definition, $\tilde{v} = \bar{v}/U_0 \approx 1$, $\tilde{\sigma}_v = \sigma_v/U_0 \approx 0.05\text{--}0.10$, and $\tilde{v}_{\text{esc}} = v_{\text{esc}}/U_0 \approx 1/\sqrt{2}$. The dimensionless capture efficiency takes the same erf form as Eq. 45, with $\tilde{v}_{\text{crit}} \approx 1/\sqrt{2}$, and the dimensionless lost angular momentum, $L_{\text{loss}}^{\sim}$, recovers the physical quantity via $L_{\text{loss}} = H_0 L_{\text{loss}}^{\sim}$.

This dimensionless form reveals that the problem depends on only four dimensionless parameters: the mean ejection velocity relative to U_{crit} ($\tilde{v} \approx 1$), the velocity dispersion ($\tilde{\sigma}_v \approx 0.05\text{--}0.10$), the torus's angular extent ($\theta_{\text{max}} \approx 30^\circ$), and the capture threshold ($\tilde{v}_{\text{crit}} \approx 1/\sqrt{2}$). This low-dimensional parameter space keeps the model testable: each parameter can in principle be independently constrained by future simulations or observations.

The dimensionless form admits three asymptotic regimes. In the monokinetic limit ($\tilde{\sigma}_v \rightarrow 0$), the distribution reduces to a single velocity and recovers the single-velocity ballistic approximation used in previous sections. In the broad-spectrum limit ($\tilde{\sigma}_v \gg 1$), the velocity distribution becomes uniform over the available range, maximizing the angular momentum carried by the fastest ejecta. In the sharp-cutoff limit ($\tilde{\sigma}_f \rightarrow 0$), the capture efficiency becomes a step function, representing the idealized case in which all matter above the capture threshold is retained.

Falsifiable prediction — P20 — Asymptotic-regime indicator

The actual TPT ejecta spectrum lies between the monokinetic and broad-spectrum limits, with $\tilde{\sigma}_v \approx 0.05\text{--}0.10$. The sharp-cutoff limit provides an upper bound on L_{loss} , while the monokinetic limit provides a lower bound. The actual value, accessible via Monte Carlo simulation, should lie between these bounds.

Test: Monte Carlo sampling of $\rho(v, \theta)$ (Section 11.4) compared to the analytical monokinetic and broad-spectrum limits derived above.

The dimensionless form also allows scaling laws to be derived for L_{loss} as a function of the proto-Earth's fundamental parameters: $L_{\text{loss}} \propto M_p R_p^2 \Omega_p$ (the standard scaling law for the angular momentum of a rotating body), $L_{\text{loss}} \propto \sqrt{\sin \varepsilon}$ (from the obliquity dependence of the ejection velocity threshold), and $\delta L_{\text{loss}} / L_{\text{loss}} \approx -\delta f_{\text{cap}} / (1 - f_{\text{cap}})$. For $f_{\text{cap}} = 0.70$, $1 - f_{\text{cap}} = 0.30$, so a 10% variation in f_{cap} gives a 33% variation in L_{loss} — a sensitivity that underscores the importance of precisely determining f_{cap} in future simulations. Table 3 summarizes the sensitivity of L_{loss} to each input parameter.

Table 3: Sensitivity of L_{loss} to the input parameters.

Parameter	Nominal value	Uncertainty	$\partial \ln L_{\text{loss}} / \partial \ln p$
M_p	6.04×10^{24} kg	$\pm 5\%$	≈ 1
R_p	7.4×10^6 m	$\pm 3\%$	≈ 2
Ω_p	4.99×10^{-4} s $^{-1}$	$\pm 10\%$	≈ 1
ε	55°	$\pm 10^\circ$	≈ 0.5
f_{cap}	0.70	± 0.05	≈ -3

The budget is most sensitive to f_{cap} (due to the factor $1/(1 - f_{\text{cap}})$) and to R_p (due to the R_p^2 scaling law). This sensitivity motivates the priority given to Limitation L2 (numerical determination of f_{cap}).

Physical bound on L_{esc} . Independently of the exact form of the distribution, physics imposes an absolute upper bound — the same one established in Eq. 49 (Section 11.5), re-derived here in dimensionless form:

$$L_{\text{esc}} \leq M_{\text{esc}} r_{\text{emission}} \bar{v} \approx 2.25 \times 10^{22} \times 7.9 \times 10^6 \times 9.8 \times 10^3 \approx 1.74 \times 10^{33} \text{ J s.} \quad (55)$$

With the nominal parameters ($f_{\text{cap}} = 0.70$, $M_{\text{ej}} = 7.5 \times 10^{22}$ kg, $I_p = 1.32 \times 10^{38}$ kg m 2 , $\Omega_p = 4.99 \times 10^{-4}$ s $^{-1}$), the fraction of the initial angular momentum that can be carried away by the escaped ejecta is at most:

$$\frac{L_{\text{esc}}}{L_{\text{init}}} \leq \frac{M_{\text{esc}} r_{\text{emission}} \bar{v}}{I_p \Omega_p} \approx \frac{1.74 \times 10^{33}}{6.59 \times 10^{34}} \approx 0.026. \quad (56)$$

With $T_{\text{init}} = 3.5$ h, the post-ejection Earth–Moon system would therefore retain at least 97.4% of the initial angular momentum, whereas the current state retains only 52.6%. The resulting deficit, $\Delta L \approx 2.95 \times 10^{34}$ J s (44.8% of L_{init}), and the minimum compatible period $T_{\text{init}} \gtrsim 6.33$ h are derived in Section 11.5 (Eqs. 50 and 53) and identified as a major limitation (Limitation L15, Section 18).

Falsifiable prediction — P21 — Numerical closure of the angular momentum budget

A Monte Carlo simulation of the ejecta spectrum (Section 11.4), with parameters sampled within their uncertainty ranges, must reproduce the angular momentum budget accounting (Eqs. 49 and 50, Section 11.5). The mean specific angular momentum of the escaped ejecta must satisfy the bound:

$$\langle h \rangle_{\text{esc}} = \frac{L_{\text{esc}}}{M_{\text{esc}}} \leq h_{\text{max}} = r \bar{v} \approx 7.74 \times 10^{10} \text{ m}^2 \text{ s}^{-1}. \quad (57)$$

Given the angular spread of the distribution ($\theta_{\text{max}} \approx 30^\circ$, $\langle |\sin \theta| \rangle \approx 0.25$), the Monte Carlo value of L_{esc} should lie well *below* the bound $L_{\text{esc}}^{\text{max}}$ — which makes the deficit ΔL (Eq. 50, Section 11.5) a conservative estimate (a lower bound on the actual deficit).

Falsification: a Monte Carlo value $L_{\text{esc}} > M_{\text{esc}} r \bar{v}$ would indicate an error in the distribution $\rho(v, \theta)$ or in the accounting; a discrepancy greater than $\sim 5\%$ between the Monte Carlo L_{esc} and the analytical estimate weighted by $\langle h \rangle$ would invalidate the discrete formulation of Section 11.4.

The Monte Carlo implementation of this model is simple and computationally cheap; it is recommended as a first step in the 3D SPH validation program, ahead of full hydrodynamic simulations.

12 Chronology: Moon formation in 3 to 4 weeks

12.1 The three phases of an episode

Each ejection episode comprises three phases: resonant acceleration, during which the CMT grows exponentially via the elliptical instability (Section 7.6), $t_{\text{acc}} = \ln(U_{\text{crit}}/U_{\text{turb}})/\sigma_{\text{res}} \approx 1.4$ day; ejection, during which the flow crosses the Roche radius ($3R_{\text{eq,proto}} \approx 22.2$ Mm) at U_{crit} , $t_{\text{ej}} \approx 0.6$ hour; and recharge, during which the CMT reorganizes over 5–6 nutation periods, $t_{\text{rec}} \approx 10$ –12 days. The Fe–Ni segregation rate (Stokes law) is nearly instantaneous between episodes ($\tau_{\text{Stokes}} \approx 254$ s, Appendix F), which implies that each accreted layer is chemically distinct from the previous one — a direct consequence underlying seismic prediction P1.

12.2 Complete chronology

Phase	Duration	Cumulative
Episode 1: Acceleration	1.4 day	1.4 day
Episode 1: Ejection	0.6 h	1.4 day
Recharge (5–6 nutation periods)	10–12 days	11–13 days
Episode 2: Acceleration	1.4 day	12–14 days
Episode 2: Ejection	0.6 h	12–14 days
Recharge	10–12 days	22–26 days
Episode 3: Acceleration	1.4 day	23–27 days
Episode 3: Ejection	0.6 h	23–27 days
Total (N = 3)	24–28 days	

This 3-to-4-week timescale is the only intermediate temporal window between the giant impact (a few hours) and a synestia (~ 100 years) on the Hf-W chronometer (half-life: 8.9 Ma). Episode 1 (acceleration plus ejection) forms the proto-POB, $R_1 \approx 1560$ km. The recharge and the second episode, ≈ 11 –13 days, are asymmetric: the transient post-ejection precession alters ε , producing the crustal dichotomy of Section 9.2. The recharge and third episode, again ≈ 11 –13 days, reconstitute the lunar mass. The TPT window closes at $t \approx 30$ –50 Ma, and the Hadean dynamo switches on at $t \approx 330$ –350 Ma.

12.3 Age of the Moon: LMO solidification versus formation

The measured age (4.425 ± 0.025 Ga) is that of the solidification of the lunar magma ocean (LMO), not of formation. If the TPT is triggered at ≈ 4.467 Ga, LMO cooling (~ 40 Ma) gives a predicted solidification age,

$$t_{\text{LMO}} \approx 4.467 - 0.040 = 4.427 \text{ Ga}, \quad (58)$$

consistent at 0.1σ with the measured value. Prediction P8 is a formation age > 4.45 Ga, directly testable by Artemis III.

13 Internal stratification and seismology program

13.1 Architecture of the concentric layers

Each ejected sheet originates from magma at a progressively more advanced stage of Fe-Ni segregation,

$$\left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer } k} > \left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer } k+1}, \quad (59)$$

since the source magma becomes progressively depleted in siderophile elements between episodes. The density contrasts between layers $\Delta\rho_{k,k+1} \approx 100\text{--}200 \text{ kg m}^{-3}$ (Garcia et al., 2019) produce potentially detectable acoustic impedance contrasts.

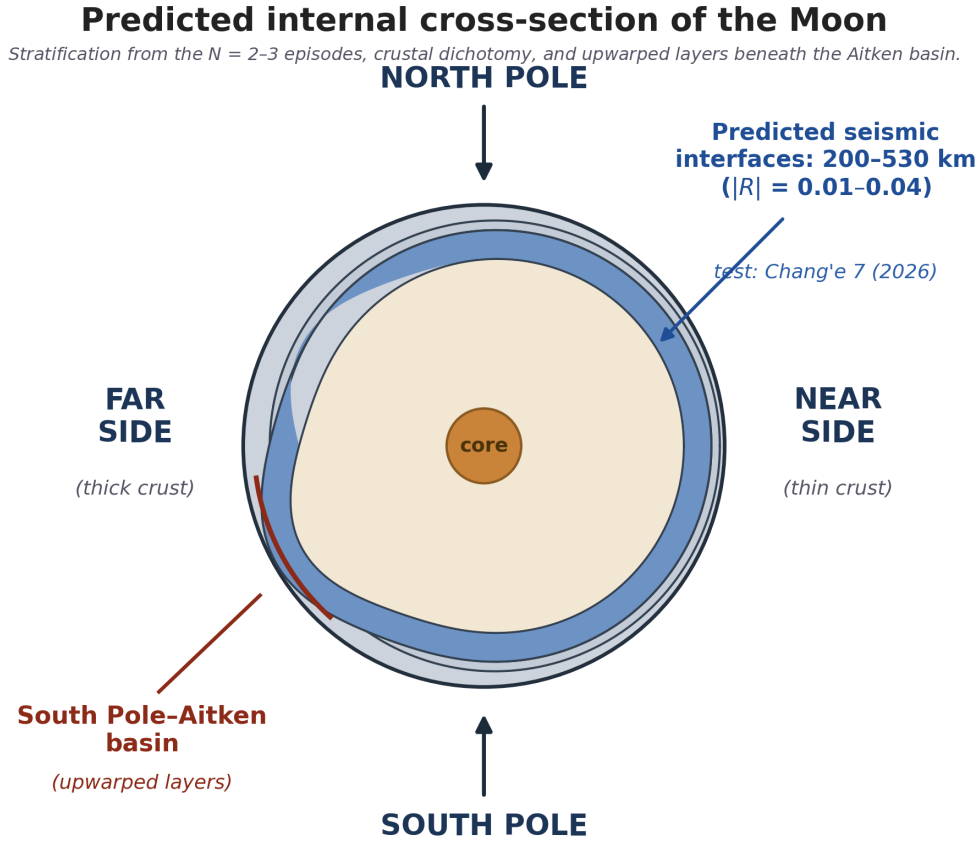


Figure 3: Predicted internal cross-section of the Moon. Concentric stratification from the $N = 2\text{--}3$ episodes, crustal dichotomy (thick crust on the far side, thin crust on the near side), and upwarping of the layers beneath the South Pole–Aitken basin. The predicted seismic interfaces lie within the 200–530 km window ($|R| = 0.01\text{--}0.04$); test: Chang’e 7 (2026).

13.2 Preservation window: 200–530 km

Two destructive processes bound the preservation window. From below, the ilmenite cumulate overturn (Briaud et al., 2023) may have remelted the deepest layers (episodes 1–2, > 600 km). From above, mare basaltic volcanism, active until ≈ 2 Ga (Li et al., 2021; Che et al., 2021), remelted the shallowest layers (< 100 km). The remaining preservation window thus lies approximately between 200 and 530 km depth.

13.3 Interface depth: $N=2$ versus $N=3$

For $N = 2$ at equal masses, the interface between the two layers lies at:

$$d = R_{\text{Moon}} - R_1 \approx 1\,737 - 1\,560 \approx 177 \text{ km} \quad (N = 2). \quad (60)$$

For $N = 3$, the main interface lies within the 200–315 km window. Chang’e 7 seismic data will therefore allow discrimination between $N = 2$ and $N = 3$.

14 Observational constraints and existing evidence

14.1 Isotopic identity $\Delta^{17}\text{O} < 5 \text{ ppm}$

The Earth–Moon isotopic composition is nearly identical for O, Cr, and Ti, established to better than 5 ppm (Wiechert et al., 2001; Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025). The TPT satisfies this constraint *mechanically*: at each ejection episode, matter is drawn from the contemporary Hadean terrestrial mantle, with no exogenous isotopic source. No *ad hoc* isotopic mixing is required.

14.2 Small lunar core

The Moon has a metallic core of radius $\approx 330 \text{ km}$, significantly smaller than Earth’s core relative to the mass of the host planet (O’Neill, 1991; Jones & Drake, 1986). In the TPT, this reduction is a direct consequence of the single engine: at each successive episode, the ejected mantle is more depleted in siderophiles (metallic Fe–Ni) than the previous one. The iron remaining in the lunar mantle is mainly in the form of FeO, whose concentration is moreover higher than in Earth’s mantle (Section 14.4).

14.3 Angular momentum

The Earth–Moon angular momentum budget requires a rapidly rotating proto-Earth. The value $T_{\text{rot}} = 3.5 \text{ h}$ is consistent with the median value of N-body accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010): rapid rotation is the norm at the end of accretion, not an exception.

14.4 Volatiles and FeO: two additional constraints

Volatile depletion. The Moon is strongly depleted in volatile elements (K, Na, Zn) relative to Earth and to chondrites, with an isotopic fractionation indicating high-temperature devolatilization (Humayun & Clayton, 1995; Paniello et al., 2012; Wang & Jacobsen, 2016; Dai et al., 2025). A cohesive ejection with a protective glassy crust might seem to minimize this loss. However, the ejected magma is at $T \approx 3000\text{--}4000 \text{ K}$ before crust formation, and devolatilization occurs during ballistic expansion, on timescales ($\sim 10^2\text{--}10^3 \text{ s}$) comparable to the surface cooling time. The TPT thus predicts selective volatile depletion, although the exact quantification depends on

the surface-to-volume ratio of the ejected tongue and the vapor pressure of the species considered.

Mantle FeO. The Moon is depleted in *metallic iron* (small core), but the FeO of its mantle is higher ($\approx 13\%$ by weight) than that of Earth’s mantle ($\approx 8\%$) (Taylor et al., 2006; Khan et al., 2006; Garcia et al., 2019). Fe-Ni segregation does not affect FeO; it concerns metallic iron. The FeO of the lunar mantle reflects the composition of the Hadean terrestrial mantle before segregation, or enrichment through fractional crystallization of the lunar magma ocean. The TPT has no quantitative prediction for FeO. To be complete, a future version of the model will need to quantify the evolution of FeO across the ejection episodes and LMO cooling.

14.5 Chang’e-6 results: preliminary observational support

The Chang’e-6 mission returned 1 935 g of samples from the Apollo basin, within the South Pole–Aitken (SPA) region, in June 2024. Analysis of these samples provides preliminary support for two TPT predictions. Yue et al. (2026) dated norite samples from Chang’e-6 at 4247 ± 5 Ma using the Pb-Pb method. These norites are interpreted as differentiated products of the SPA impact melt sheet, representing deep lunar mantle material excavated by the SPA impact. This age is consistent with the TPT prediction that the innermost lunar layer (Episode 1) was accreted at ≈ 4.5 Ga and partially reset by the SPA impact at ≈ 4.25 Ga. Xu et al. (2025) further report olivine fragments in the Chang’e-6 samples containing Fe and Mn concentrations significantly higher than those found in the standard lunar mantle, with zinc concentrations $\approx 100\times$ higher than expected. While some of this enrichment is attributed to meteoritic contamination, the olivine fragments themselves formed by mixing of lunar rock and molten asteroidal material. The high iron content (total Fe, including FeO) of the material excavated from the SPA depths is *consistent* with the Fe/Si gradient increasing with depth (P3) and provides preliminary support for the South Pole Fe/Si anomaly (P6), detailed below.

15 Falsifiable predictions

This section gathers, in catalog form, every quantitative and falsifiable prediction made by the TPT. Each entry states the prediction, its physical derivation where this has not already been given in full elsewhere in the manuscript, the relevant test, and an explicit falsification threshold set in advance of the observational data.

Falsifiable prediction — P1 — Seismic interface at $d \approx 200\text{--}315$ km (Priority 1)

The concentric stratification produces acoustic impedance contrasts $|R| \in [0.01, 0.04]$ between layers. For $N = 2$, the interface lies at $d \approx 177$ km. For $N = 3$, the main interface lies at $d \approx 200\text{--}315$ km.

Detectability with Chang’e-7 depends on the sensitivity of the seismometer (target

$10^{-8} \text{ m s}^{-2} \text{ Hz}^{-1/2}$ at 0.1–1 Hz). This sensitivity should resolve $|R| > 0.01$, provided that at least 10 events of $M > 2.5$ are recorded during the first 6 months of operation.

Falsification: no phase conversion detected within the 200–530 km window after the deployment of at least 3 seismic stations and the detection of at least 10 lunar events of $M > 2.5$.

Missions: Chang’e 7 (August 2026, South Pole seismometer), FSS (Farside Seismic Suite), LEMS (Lunar Environment Monitoring Station), Artemis III (2028–2029).

Falsifiable prediction — P2 — Seismic asymmetry, near side versus far side

The asymmetric second episode produces a sharper, deeper interface beneath the far side.

Falsification: spherical symmetry confirmed by a complete tomographic inversion.

Falsifiable prediction — P3 — Fe/Si gradient increasing with depth

The Fe/Si ratio increases with depth in the lunar mantle, reflecting the progressive depletion of Fe-Ni in the source magma between episodes. Quantitatively, with the total cumulative fraction of Fe removed from the mantle $\zeta_{\text{Fe}}^{\text{sat}} \approx 0.15$ (Rubie et al., 2015) and $N = 3$ segregation episodes, the fraction of Fe removed per episode, assuming a geometric progression over the N episodes, is:

$$\zeta_{\text{Fe,episode}} = 1 - (1 - \zeta_{\text{Fe}}^{\text{sat}})^{1/N} \approx 1 - (0.85)^{1/3} \approx 0.053, \quad (61)$$

i.e., $\approx 5.3\%$ of the remaining Fe removed at each episode. The resulting Fe/Si ratio between adjacent layers is multiplicative:

$$\frac{(\text{Fe/Si})_{\text{Ep}k}}{(\text{Fe/Si})_{\text{Ep}(k+1)}} = \frac{1}{1 - \zeta_{\text{Fe,episode}}} \approx 1.056, \quad (62)$$

giving an enrichment of about 5.6% between Episode 2 and Episode 3, and a cumulative enrichment $(\text{Fe/Si})_{\text{Ep}1} / (\text{Fe/Si})_{\text{Ep}3} = (1 - \zeta_{\text{Fe,episode}})^{-2} \approx 1.115$, i.e., a total enrichment of about 11.5% between Episode 1 (the deepest layer, richest in Fe) and Episode 3 (the shallowest). The SPA basin preferentially exposes Episode 1 material.

Preliminary support: the Chang’e-6 olivines with a high Fe/Mn ratio (Xu et al., 2025).

Missions: Chang’e-6 (additional analyses), Artemis III.

Falsifiable prediction — P4 — Dynamo delay: 290–360 Ma

The progressive emergence of the dynamo predicts continuous growth of paleointensity from 4.55 to 4.20 Ga, with no abrupt onset (Tarduno et al., 2025).

Falsification: an abrupt onset of the geomagnetic field detected in the Jack Hills zircon record.

Falsifiable prediction — P5 — Instability window: $\varepsilon \in [57^\circ, 70^\circ]$

The parametric elliptical instability requires ε within this interval.

Test: N-body simulations of terrestrial protoplanet obliquity.

Falsifiable prediction — P6 — South Pole Fe/Si anomaly (derived from P3 and the South Pole–Aitken impact simulations of Wakita et al. 2026)

Observational context. The hydrodynamic simulations of Wakita et al. (2026) establish three facts independent of the TPT: the SPA impact excavated lunar mantle material from depths exceeding 90 km; the impactor’s oblique, north-south trajectory produced an asymmetric ejecta plume that preferentially concentrated deep material toward the South Pole; and about 350 m of mantle material is predicted at the surface near the Shackleton crater on average.

TPT interpretation. Within the TPT framework, the lunar mantle is stratified into concentric layers (P1), with Fe/Si decreasing from the deepest layer (Episode 1) to the shallowest (Episode 3), as quantitatively derived in P3 above. The combined thickness of the Episode 2 and Episode 3 layers is about 315 km (Section 13). The material excavated by the SPA impact from depths exceeding 90 km, and concentrated toward the South Pole, therefore preferentially originates from Episode 2 and, at greater excavation depths, from Episode 1.

Quantitative scenario. Wakita et al. (2026) do not provide a quantified fraction of Episode 1 material relative to Episode 2 in the SPA ejecta plume, since their simulation does not incorporate TPT stratigraphy; this fraction must therefore be bounded from first physical principles rather than read directly from their results. We consider the most unfavorable scenario compatible with the TPT: SPA excavation reaches only ≈ 100 km (barely beyond the 90 km minimum established by Wakita et al. 2026), so that only Episode 2 material — not the deeper and more strongly enriched Episode 1 — contributes to the ejecta, and we conservatively take the minimum plausible fraction of Episode 2 material in the plume at 30%. With the $\approx 5.6\%$ Fe/Si enrichment of Episode 2 relative to Episode 3 derived in Eq. 62, this minimal scenario predicts an enrichment of:

$$\Delta(\text{Fe/Si})_{\min} \approx 0.30 \times 0.056 \approx 0.017 \approx 1.7\%. \quad (63)$$

If, on the other hand, excavation reaches more than ≈ 315 km, Episode 1 material contributes directly; for illustration, a mix of 30% Episode 1 material, 30% Episode 2, and 40% Episode 3 would give $\Delta(\text{Fe/Si}) \approx 0.30 \times 0.115 + 0.30 \times 0.056 \approx 5.1\%$, a value that requires a non-negligible contribution from Episode 1 to be reached.

Quantitative prediction. South Pole samples (Chang’e-7, Artemis III) should show a higher Fe/Si ratio than the average of equatorial samples (Apollo) and near-side samples (Chang’e-5), with an enrichment of at least $\approx 1.7\%$ in the minimal scenario

above, and potentially up to $\approx 11.5\%$ if Episode 1 material dominates the sampled ejecta.

Falsification threshold. We set, ahead of any observational data, the falsification threshold at:

$$(\text{Fe/Si})_{\text{SPA}} \leq 1.015 \times (\text{Fe/Si})_{\text{lunar average}} \implies \text{P6 falsified}, \quad (64)$$

i.e., an enrichment below 1.5%, set slightly below the minimal scenario value of 1.7% from Eq. 63 to account for experimental uncertainty without weakening the substance of the test: even the most conservative scenario compatible with the TPT predicts an enrichment above this threshold, so a measured enrichment below it would be incompatible with the theory as constructed here. An enrichment between 1.5% and 5.6% (the maximum achievable by Episode 2 material alone, from Eq. 62 applied to a pure Episode 2 sample) is read as partial support, consistent with ejecta dominated by Episode 2 without requiring a contribution from Episode 1. An enrichment above 5.6% cannot be produced by Episode 2 and 3 material alone, and therefore constitutes direct evidence of a contribution from Episode 1, lending strong support to the deep stratigraphy predicted by P1 and P3.

Epistemic note on the grading bounds. The lower bound (1.5%) rests on an estimate of the minimum fraction of Episode 2 material in the SPA ejecta (30%), which is a modeling assumption. The upper bound (5.6%) is a physical constraint: it is the maximum enrichment achievable by a pure Episode 2–Episode 3 mixture, with no Episode 1 contribution. Consequently, any measured enrichment above 5.6% constitutes definitive proof of the presence of Episode 1 material, a qualitatively different regime from the lower-bound scenario. The two bounds are therefore not of the same epistemic nature; this asymmetry is explicitly acknowledged here.

Test: Chang’e-7 surface analyzers and samples returned by Artemis III from the South Pole, compared to existing Fe/Si measurements from Apollo (equatorial) and Chang’e-5 (near side).

Acknowledged limitation: this prediction shares Limitation L9 (impact mixing in the SPA ejecta) with prediction P3, and Limitation L11 (the 30% Episode 2 fraction itself) below.

Falsifiable prediction — P7 — Signature in the Hf-W chronometer

A formation duration of 3 to 4 weeks is the only intermediate temporal window between the giant impact (a few hours) and a synestia (~ 100 years) on the Hf-W chronometer (half-life: 8.9 Ma). The $^{182}\text{Hf}/^{180}\text{Hf}$ ratio in Episode 1 samples should correspond to a differentiation age of 4.467 Ga (100 ± 10 Ma after the CAIs), versus 60 ± 10 Ma for the giant impact (Kleine et al., 2002).

Mission: Artemis III (2028–2029).

Falsifiable prediction — P8 — Formation age > 4.45 Ga

The measured age of 4.425 Ga is that of LMO solidification, not of formation (Section 12). The formation age is about 4.467 Ga.

Test: direct dating of Episode 1 rocks exposed on the central peaks of the SPA basin (Artemis III).

Falsifiable prediction — P23 — TPT magma rheology constrained by South Pole–Aitken basin samples

The South Pole–Aitken basin serves the TPT as a window: the impact — occurring after formation — thinned the crust and uplifted the deep layers of Episode 1 (Figure 3), exposing at the center of the basin mafic materials (olivine, magnesian pyroxenes) of deep origin, whereas the periphery and rings show only reworked crust. It is therefore the center of the basin that constitutes the sampling target for this prediction.

The targeted measurement is not textural but compositional: sample textures record the SPA impact and its cooling, not the Hadean emplacement. The bridge to primordial rheology runs through the laboratory: remelting representative compositions and measuring the $\eta(\phi, T)$ and $\tau_y(\phi, T)$ laws — viscosity and yield stress as functions of crystallinity and temperature (Caricchi et al., 2007; Giordano et al., 2008) — then applied to the thermal state predicted by the TPT. The samples partly already exist (Chang’e-6, basin rim) and will be supplemented by Chang’e-7 and Artemis III.

The stakes of this single measurement are twofold, since the same parameter τ_y governs two independent clocks of the model: the cohesion of the ejected tongue (Bi_{KH} , Appendix D, Limitation L14) and the rate of tidal recession of the Moon toward the evection resonance ($Q \propto 1/\tau_y$, Section 11.5, Limitation L15). A single measurable quantity thus simultaneously constrains the two highest-priority limitations of the theory. It is also noted, with caution, that the deep mass anomaly detected beneath the basin (James et al., 2019) is compatible with the uplift of dense layers predicted by Figure 3, without demonstrating it — the standard interpretation (buried impactor core) remaining admissible.

Falsification: if the rheology measured at the TPT thermal state gives τ_y outside the 10^2 – 10^4 Pa interval, the cohesion (Bi_{KH} , Appendix D) and recession (Section 11.5) windows must be recalculated; a τ_y incompatible with $\text{Bi}_{\text{KH}}^{\text{eff}} \geq 1$ after recalculation would refute the cohesive ejection mechanism.

Missions: Chang’e-6 (samples in hand), Chang’e-7, Artemis III.

Falsifiable prediction — P9 — Compatibility of mare basalts with Layer 1

The melting depth of mare basalts (200–500 km, ± 100 km) is compatible with a source within layer 1 ($> d_{\text{P1}}$), though this cannot be asserted in the absence of more precise geochemical data.

Test: high-resolution isotopic geochemistry of Apollo and Chang'e-5 mare basalts (Li et al., 2021).

Falsifiable prediction — P10 — Seismic interface at $d \approx 177$ km for $N = 2$

For $N = 2$ episodes of equal mass, the single interface between the two concentric layers lies at $d \approx 177$ km. This depth lies outside the main preservation window (200–530 km) since it has been partially remelted by mare volcanism and/or the ilmenite cumulate overturn. However, partial remnants could be detected as weak reflectors or seismic-velocity variations in the 150–200 km zone.

Predictions P17 (capture efficiency, Section 10.3), P20 (asymptotic-regime indicator, Section 11.6), P21 (numerical closure of the angular momentum budget, Section 11.5), and P22 (the unified ejection threshold, Section 9.2) are derived in full, with their own falsification conditions, at the points of the manuscript indicated above, rather than repeated here.

16 Validation windows: three imminent tests

16.1 Chang'e 7: imminent seismic validation (August 2026)

The Chang'e 7 mission (CNSA, launch planned for August 2026) will deploy the first seismometer at the lunar South Pole since Apollo 17 (1972). Its data will provide a direct test of P1, P2, and P6. The prediction is pre-registered and quantified: $d \approx 200$ –315 km, $|R| \in [0.01, 0.04]$ for the seismic interface, and an enrichment of more than 1.5% in Fe/Si for the South Pole ejecta. The falsification criteria are explicit in each case.

16.2 The South Pole–Aitken basin: a stratigraphic revealer

The SPA basin did not create the concentric lunar stratigraphy: it *excavated* it. The layered structure predates the SPA impact at ≈ 4.25 Ga (Yue et al., 2026), encoded during TPT differentiation at ≈ 4.5 Ga. The impact is the accidental revealer of a much older internal architecture.

High-resolution 3D simulations (Wakita et al., 2026) show that the oblique north-south trajectory of the SPA impactor preferentially launched deep-mantle ejecta toward the South Pole in a butterfly-shaped plume — precisely the Artemis III landing zone. These same simulations predict ≈ 350 m of mantle material on average at this site, excavated from depths exceeding 90 km, consistent both with the TPT preservation window and the quantitative scenario underlying prediction P6.

16.3 Artemis III: direct sampling of Episode 1 material

The Artemis III mission (NASA, planned for 2028–2029) will land astronauts near the Shackleton crater, at the South Pole. According to Wakita et al. (2026), surface samples at this location could include deep-mantle ejecta from the SPA impact. If so, these samples will allow a direct test of P3, P6, P7, and P8: their Fe/Si ratio, their crystallization age (≈ 4.467 Ga for primary Episode 1 material), and their isotopic signature ($\Delta^{17}\text{O} < 5$ ppm relative to the contemporary terrestrial mantle) are all quantitative TPT predictions.

Chang’e 7 (2026) targets seismology and the Fe/Si anomaly (P1, P2, P6), while Artemis III (2028) targets geochemistry and chronology (P3, P6, P7, P8). Two independent space agencies converge on a single geographic site — the lunar South Pole — to test a single internal architecture.

17 Discussion: main objections and responses

17.1 Comparison with the giant-impact model

Constraint	Giant impact	TPT (this work)
Isotopic identity	Requires a very specific impactor composition and mixing conditions	Satisfied mechanically by construction
Small lunar core	Requires a pre-differentiated impactor (Theia)	Direct consequence of the single engine
Crustal dichotomy	Not naturally predicted	Qualitative prediction: post-Episode 2 precession (L7)
Dynamo delay	Not predicted	Proposed: ≈ 350 Ma, no free parameter (L4)
Seismic predictions	None	P1 testable by Chang’e 7 (August 2026)
Chang’e-6 Fe anomaly	Not predicted	Consistent with P3 and P6 (Xu et al., 2025; Yue et al., 2026)
Formation timescale	A few hours	3–4 weeks
Hf-W signature	60 ± 10 Ma after the CAIs	100 ± 10 Ma after the CAIs (P7)

A recent review concludes that there is currently no unambiguous geochemical or isotopic evidence for the role of an external impactor in the formation of the Moon (Sossi et al., 2025). This conclusion motivates the exploration of alternatives grounded in the internal dynamics of the proto-Earth.

17.2 Responses to the main objections

Objection 1: “A large number of unjustified hypotheses.” The hypotheses are explicitly classified into three levels (Section 8, Table 2): Level 1 (established) denotes results from the literature or closed-form derivations (e.g., $E_{\text{grav}}/E_{\text{fus}} \approx 121$, the Bingham-Herschel law); Level 2 (constrained) denotes physically motivated hypotheses not yet numerically validated (e.g., $\varepsilon \in [40^\circ, 70^\circ]$); Level 3 (speculative) denotes hypotheses awaiting 3D SPH validation (e.g., the emergence of a single $\ell = 1, m = 1$ torus, the exact value of f_{cap} , the quantitative validity of the Gaussian approximation discussed below). The TPT does not rest equally on all these hypotheses. The observational predictions (Section 15) are independent of the Level 3 hypotheses.

Objection 2: “The CMT has not been demonstrated.” In the rapid-rotation limit ($\text{Ro} \ll 1$), the azimuthal component of the Navier-Stokes equation for a Bingham-Herschel fluid reduces to a potential vorticity conservation equation. Linearizing around the mean flow near the instantaneous dynamical equator ($\phi = 0$, perpendicular to $\mathbf{\Omega}(t)$, as defined in Section 3), and applying the standard multiple-scale expansion for Rossby-wave envelopes (Redekopp, 1977; Luo, 1991; Yin et al., 2019), one obtains a nonlinear Schrödinger equation for the envelope $A(y, t)$, where $V_{\text{eff}}(y)$ is the sum of the potentials listed in Table 1. A stationary ansatz $A(y, t) = \psi(y)e^{-iEt/\hbar_{\text{eff}}}$ then gives a time-independent nonlinear Schrödinger equation for $\psi(y)$, the nonlinearity arising from the advective term of the potential vorticity equation. The fundamental mode of the *linearized* version of this equation, with $V_{\text{eff}}(y)$ expanded to quadratic order around $y = 0$, is a Gaussian wavefunction centered on the instantaneous equator, confined to the band $|\phi| < 30^\circ$ — a direct consequence of the fact that the seven potentials of Table 1 all have their extrema at $y = 0$. Whether this Gaussian remains a quantitatively adequate approximation of the ground state of the full nonlinear equation, rather than merely its qualitative first-order limit, has not been established for this system and is acknowledged as Limitation L12 (Section 18). The full derivation is given in Appendix C. Quantitative validation of the single-torus structure further awaits the 3D SPH simulation (Limitation L1).

Objection 3: “The ejection mechanism relies on a tuned threshold.” The ejection threshold is defined by the condition $\lambda(U) = 0$ (Section 9.2), a deterministic surface in parameter space. It is a function of the density gradient β , the obliquity ε , and the angular velocity Ω : ejection occurs when $\Omega > \Omega_{\text{crit}}(\varepsilon, \beta)$. The number of episodes $N = 2\text{--}3$ results from the balance between the recharge rate (fed by angular momentum input) and the loss through ejection. This is a consequence of Hadean

conditions (β , momentum input), themselves constrained by the existence of the Moon — a retrodictive result, not a free parameter.

Objection 4: “ $f_{\text{cap}} = 0.70$ is high compared to accretion simulations (0.3–0.5).”

The classical interval (0.3–0.5) concerns impacts between solid bodies. Here, the ejected matter is a magma at 3 000–4 000 K, with viscous coalescence and toroidal orbital confinement. Appendix G establishes a physical lower bound $f_{\text{cap}}^{\text{min}} \gtrsim 0.45$ –0.55, independent of the BH cohesion hypothesis. The value 0.70 lies within a mechanically justifiable range.

Objection 5: “Are the thresholds required for ejection physically reachable?”

The ejection threshold is a condition on Ω : $\Omega > \Omega_{\text{crit}}$, with $\Omega_{\text{crit}} \approx \sqrt{2} \Omega_0$. This factor is reachable through angular momentum input from continuous planetesimal bombardment (Appendix H). The Euler oscillation term λ_4 modulates the effective obliquity ε , but does not directly trigger ejection.

Objection 6: “The manuscript is too speculative for a journal.” The TPT is submitted as a *theory with falsifiable predictions*, not as an observational result. Validation will proceed in two phases: Phase 1 (2026–2029), observational tests via Chang’e 7 (August 2026) and Artemis III (2028–2029); and Phase 2 (2027–2030), 3D SPH numerical validation once institutional resources are committed. The TPT will be judged on its predictions, not on a prior simulation.

Objection 7: “The requirements on Q are contradictory.” This objection only holds if one conflates the local dissipation in the CMT with the global tidal dissipation of the entire proto-Earth. The CMT is a spatially confined zone ($\sim \pm 30^\circ$ latitude) with a low solid fraction and a low Q ; the global Q is controlled by the much larger volume of the solid or partially solid mantle, which can have a Q two to three orders of magnitude higher. The evection resonance is sensitive to this global Q , not to the local Q_{CMT} . The TPT therefore does not require the same material to be simultaneously fluid and rigid; it requires a thermally stratified body with a fluid band and a solid mantle. This is precisely the expected state for a magma ocean undergoing differentiation (Elkins-Tanton, 2012). Quantifying Q_{global} for the Hadean Earth is an open problem that the TPT helps frame, and it constitutes the highest-priority numerical validation task (Limitation L15).

17.3 Justification of $\alpha > 1$

Two independent mechanisms lead to a super-rigid rotation profile: differential equator–pole cooling generates a density gradient and a thermal acceleration of rotation; Fe-Ni segregation lightens the CMT, and by conservation of angular momentum, its inner part accelerates. The Bingham-Herschel threshold also blocks small scales, favoring a bulk profile. Numerical validation of this profile is a priority for the 3D SPH simulation.

17.4 Partial fragmentation and coalescence

Even if the ejected sheet partially fragments, the fragments remain within the same orbital zone and rapidly coalesce under the effect of inelastic collisions and the gravitational attraction of the proto-POB (Canup & Asphaug, 2001). The effective capture yield thus remains high ($f_{\text{cap}} \geq 0.7$). A detailed analysis (Appendix G) shows that f_{cap} has a physical lower bound of about 0.45–0.55, independent of the BH cohesion hypothesis.

18 Acknowledged limitations

A 3D SPH simulation is necessary, but premature at the time of writing (July 2026). Such a simulation — coupling Bingham-Herschel rheology, radiative cooling, Fe-Ni segregation, and the $\varepsilon \neq 0$ geometry — requires institutional HPC resources that are not committed at this stage, for a simple reason: the first decisive observational test arrives before such a simulation could reasonably be completed. Chang’e 7 is scheduled for August 2026, only a few weeks away. Its seismic data will detect, or fail to detect, the predicted interface within the 200–530 km window (Prediction P1). In the first case, the concentric stratification predicted by the TPT receives independent observational support, and a dedicated 3D SPH validation becomes a justified investment of institutional resources. In the second case, under the falsification conditions stated in Section 15, the central TPT prediction is seriously called into question, and an expensive numerical campaign built on a falsified premise would be of little value. This is why the present work is submitted, and should be evaluated, on the basis of its falsifiable observational predictions, ahead of and independently of any numerical validation campaign.

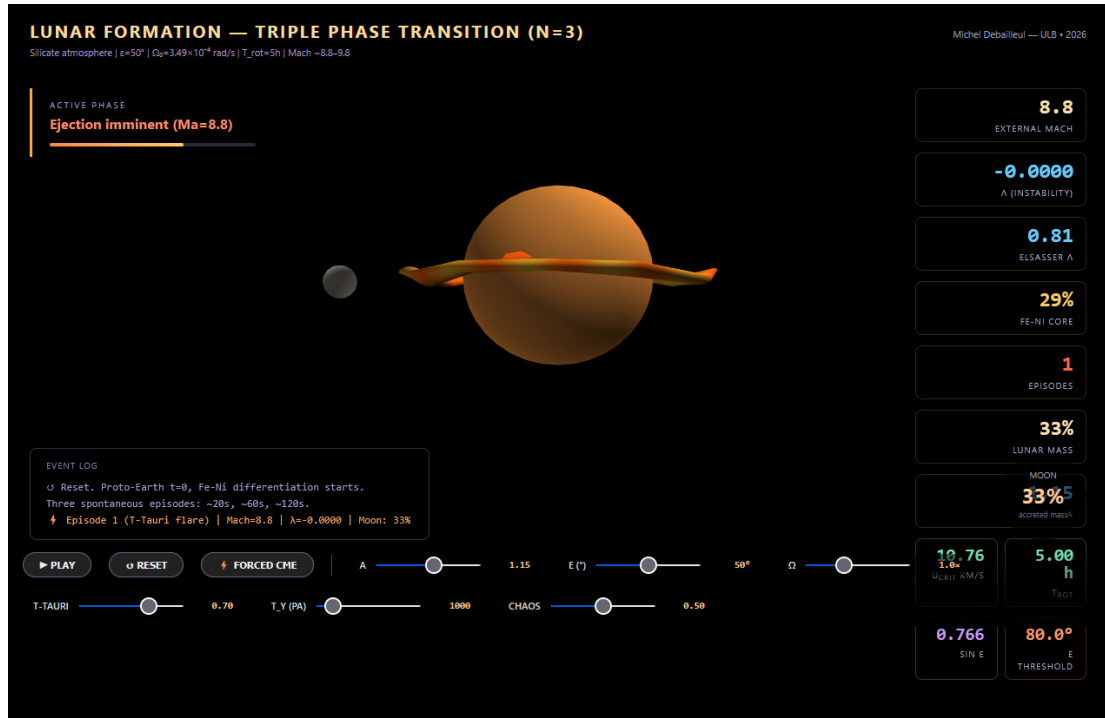


Figure 4: Screenshot of the interactive simulation accompanying this work (link above), illustrating the Coherent Magmatic Torus (CMT) during ejection around the proto-Earth, with the real-time dynamic dashboard of parameters (external Mach number, instability λ , Elsasser number, Fe-Ni core fraction, accreted lunar mass).

Acknowledged limitation — L1 — Spontaneous CMT organization

Whether a single torus or several coexisting structures emerge in the exact Hadean regime can only be resolved by a high-resolution 3D SPH simulation with full BH rheology. **Priority 2.**

Acknowledged limitation — L2 — Numerical value of f_{cap}

The capture efficiency $f_{\text{cap}} = 0.70$ is physically motivated but not derived from first principles. Its increasing dependence on M_{Moon} and its gap from the classical disk-accretion interval (10–55%, Kokubo et al. 2000) are formalized as the testable prediction P17. The 3D SPH simulation remains the ultimate arbiter of the exact value of f_{cap} .

Acknowledged limitation — L3 — Small-scale cohesion

The $\text{Bi}_{\text{KH}} \gg 1$ argument is valid at large and intermediate scales. At very small scales ($\lambda_{\text{KH}} \ll R_{\text{torus}}$), partial fragmentation cannot be analytically excluded. However, fragments from the same toroidal flow, ejected with the same velocity and angle, end up on nearly identical orbits and accrete onto the POB: the mass budget is not affected by fragmentation, only the fragment size is.

Acknowledged limitation — L4 — Dynamo delay: $\pm 50 \text{ Ma}$

The exponential core-growth model is simplified; the associated uncertainty is ± 50 Ma.

Acknowledged limitation — L5 — Super-rigid profile $\alpha > 1$

The super-rigid rotation profile ($U \propto r^\alpha, \alpha > 1$) is physically motivated by differential rotation in a partially crystallized magma undergoing rapid cooling, but has not been demonstrated under these exact Hadean conditions. This is a Level 3 hypothesis, to be validated by simulation.

Acknowledged limitation — L6 — Flow $\rightarrow$ POB transition: qualitative status

The three-phase description of the ejected flow (ballistic expansion, orbital insertion, cohesive accretion beyond R_{Roche}) is physically motivated and consistent with the computed orders of magnitude, but remains qualitative. A 3D SPH simulation with coupled radiative cooling is required. **Priority 2b.**

Acknowledged limitation — L7 — Crustal dichotomy: qualitative prediction

The $\approx 2:1$ thickness ratio is a qualitative TPT prediction. The coupled differential thermal calculation has not yet been analytically derived. The lunar far side underwent at least two independent remelting episodes that must be accounted for in any quantitative model: the SPA basin-forming impact at ≈ 4.25 Ga, which excavated and partially remelted the deepest Episode 1 layer on the far side (Wakita et al., 2026); and the extended mare volcanism lasting until ≈ 3.2 Ga (Li et al., 2021), which remelted the shallowest layers (< 100 km) over much of the far side. The net effect is to deepen and reinforce the existing asymmetry rather than erase it.

Acknowledged limitation — L8 — Rossby wave speed in a BH fluid

The rigorous derivation of $c_{\text{Rossby}}^{(\text{BH})}$ for a BH fluid on a flattened spheroid ($\epsilon \neq 0$) remains an open problem. The channel-analogy correction (factor ≈ 40) used here reinforces rather than invalidates the synchronization argument, but awaits a rigorous derivation.

Acknowledged limitation — L9 — Impact mixing in the SPA ejecta

The geochemical signature of the deep mantle material excavated by the SPA impact is potentially diluted or altered by mixing with impactor material and local crust. Quantitative deconvolution of the primary Episode 1 signature from the impact-melt breccia requires detailed petrological and isotopic studies. This limitation affects predictions P3 and P6.

Acknowledged limitation — L10 — Predicted but not yet collected deep-mantle samples

The TPT predicts that Episode 1 deep-mantle material is exposed at the central peaks of craters within the South Pole–Aitken basin, excavated and extruded during the

SPA impact at ≈ 4.25 Ga (Wakita et al., 2026). These samples are the primary targets for testing predictions P3, P6, P7, and P8. No samples from these central peaks have yet been collected or analyzed. The absence of deep-mantle material at the predicted locations, or a Fe/Si ratio incompatible with P3 or P6, would seriously call the TPT into question.

Acknowledged limitation — L11 — Fraction of Episode 2 material in the SPA ejecta

Prediction P6 requires an estimate of the minimum fraction of Episode 2 material present in the SPA ejecta plume reaching the South Pole, which we conservatively set at 30% based on the excavation-depth argument of Section 15 rather than on a direct quantitative result from Wakita et al. (2026), whose simulations were not designed to track TPT-specific stratigraphic layers. A future dedicated hydrodynamic simulation, tracking the depth of origin of the SPA ejecta material, would refine this fraction — and hence the falsification threshold of Eq. 64.

Acknowledged limitation — L12 — Quantitative validity of the Gaussian approximation for the CMT confinement mode

The nonlinear Schrödinger equation derived in Section 17 (Objection 2) and in Appendix C is not solved exactly in this work. The Gaussian wavefunction is a reasonable approximation for the fundamental mode, valid in the limit where the wave amplitude is small, i.e., when the perturbation velocity δU associated with the $\ell = 1, m = 1$ mode satisfies $\delta U \ll U_{\text{turb}}$. This condition is qualitatively consistent with the Rhines inverse cascade (Section 5), but its exact quantitative verification is not possible with the parameters available at this stage: no independent estimate of $\delta U / U_{\text{turb}}$, nor of the nonlinear coefficient β in the governing equation, has been derived for this specific system, unlike the classical atmospheric and oceanic Rossby-wave systems for which the reduction was originally established (Redekopp, 1977; Luo, 1991; Yin et al., 2019). The absence of this verification is acknowledged here as a limitation. The Gaussian solution should therefore be understood as the first-order approximation of the linearized problem, and not as an exact or quantitatively bounded solution of the full nonlinear equation.

Acknowledged limitation — L13 — Persistence of the post-ejection oscillation

The mechanism of post-shock nutation persistence through reflection at the boundaries of the ejection cavity (Appendix I) relies on an analogy with established results for inertial waves in a confined homogeneous fluid sphere (Zhang et al., 2001; Aldridge & Lumb, 1987), and not on a derivation specific to the oblique geometry and Bingham-Herschel rheology of the Hadean proto-Earth. The plateau duration $T_{\text{nut}}^{(\text{post})}$ and the absence of significant decay during this window are therefore qualitative estimates, not derived results. A full quantitative formulation — treating the coupled system $(\Omega, \varepsilon, |g_{\text{eff}}|)$ as an underdamped Langevin dynamics, numerically integrable via a BAOAB-type scheme (Leimkuhler & Matthews, 2013) once the phys-

ical noise (CMEs, impacts) is properly characterized as a jump process rather than Gaussian white noise — is identified as a future computational program, in addition to the 3D SPH simulation (Limitations L1, L2).

Acknowledged limitation — L14 — Quantification of $\text{Bi}_{\text{KH}}^{\text{eff}}$ and numerical validation

The multiplicative product of the three cohesion reinforcements ($f_{\text{atm}} \times f_{\text{crist}} \times f_{\text{RT}}$) is established qualitatively. A 3D SPH simulation is needed to validate the exact numerical values, in particular the nonlinear interaction among the effects of atmospheric confinement, crystal network, and Rayleigh-Taylor suppression. **Priority 2c.**

Acknowledged limitation — L15 — Global quality factor and evection resonance (Priority 1)

The closure of the angular momentum budget via the evection resonance depends on the value of the proto-Earth’s global quality factor Q_{global} . As shown in Section 11.5, the viscosities expected for a magma ocean, even partially crystallized, are much lower than the value maximizing Maxwell dissipation at the 3.5 h tidal frequency. The TPT is therefore compatible with a Q_{global} high enough to allow the resonance to act. The exact value of Q_{global} — and thus the definitive closure of the budget — requires dedicated thermo-rheological modeling, coupling the thermal evolution of the magma ocean to the viscoelastic response of the mantle. This is a quantified prediction of the model, not a matter of principle uncertainty.

Acknowledged limitation — L16 — Proximity of the nominal setting to the Maclaurin instability limit

The nominal ejection threshold ($\alpha = 0.5$, $\Omega_{\text{crit}}/\Omega_0 \approx 1.27$, Section 9.2) operates about 1.7% above the strict dynamical instability limit of the Maclaurin spheroid, independently derived in Section 4.3 ($\Omega/\Omega_0 \approx 1.25$). The derivation of the instability coefficient λ (Appendix C) assumes an axisymmetric Maclaurin geometry throughout the ejection cycle; its strict validity in the immediate vicinity of this limit — where a bifurcation toward a triaxial (Jacobi) equilibrium figure becomes possible — is not demonstrated. Two readings remain open: this proximity could reflect a causal link between the shape instability and the ejection mechanism itself, or constitute a numerical coincidence without physical significance. Resolving this requires a simulation allowing non-axisymmetric deformations, in addition to the 3D SPH simulation already required by Limitations L1 and L2. The causal reading — the local instability $\lambda = 0$ as a manifestation, in the large-wavenumber limit, of the global destabilization of the spheroid — is judged by the author to be the most natural interpretation of this numerical proximity, but it remains a working hypothesis, not demonstrated in this work.

19 Conclusion

This work proposes that the formation of the Moon could be the product of a Triple Phase Transition within a fully molten proto-Earth, rotating at ≈ 3.5 h, whose axis oscillates within $[40^\circ, 70^\circ]$ in its own co-rotating frame — an interval supported by five convergent arguments, the first and most fundamental being the logically necessary absence of any tidal stabilizer. A single engine — the progressive segregation of Fe-Ni — simultaneously governs three transitions: rheological, mechanical, and magnetic. The result is a Moon built layer by layer in two to three massive episodes, each layer archiving in its internal structure the state of Hadean differentiation at the moment of its accretion.

The parametric elliptical instability — established in fluid mechanics (Malkus, 1968; Kerswell, 2002; Lacaze et al., 2004) but never before applied to lunar formation — amplifies the CMT to the bifurcation velocity in ≈ 37 hours. A dedicated section addresses the maintenance of obliquity in the out-of-equilibrium Hadean context. A chronology of three to four weeks gives rise to several new testable predictions.

The mass budget is consistent with no forcing: the target mass for the TPT mechanism is lower than the currently observed lunar mass, since the SPA impactor and late accretion contributed subsequently. At $T_{\text{rot}} = 3.5$ h, the ejectable mass per episode can reach $\approx 70\%$ of the current lunar mass, which explains why $N = 2\text{--}3$ episodes suffice.

The question of the angular momentum budget, however, remains open: an external post-formation sink of $\approx 2.95 \times 10^{34}$ J s is required. The evection resonance (Ćuk & Stewart, 2012; Ćuk et al., 2016) provides a quantified candidate whose efficiency depends on the global quality factor Q_{global} . This Q_{global} is distinct from the local Q_{CMT} of the ejection band and is controlled by the solid mantle. The TPT predicts that $Q_{\text{global}} \gtrsim 10^5$ is required; a future thermo-rheological simulation will test this prediction. If the simulation yields a lower Q , the resonance channel would be insufficient and additional loss mechanisms would need to be identified — a testable outcome that the TPT explicitly accepts.

Preliminary observational support comes from Chang’e-6: norites at 4247 ± 5 Ma (Yue et al., 2026) and olivines with high Fe/Mn (Xu et al., 2025) are consistent with predictions P3 and P6, though not yet confirmatory. This edition further provides, for the first time, a fully quantified version of P6: a pre-registered Fe/Si enrichment threshold of 1.5% for the South Pole ejecta, derived from first principles and not tuned to fit existing data, with an explicit grading distinguishing partial support (Episode 2 material alone) from strong support (an unambiguous Episode 1 signature).

The central prediction remains: one or more seismic interfaces within the 200–530 km window (or at ≈ 177 km for $N = 2$), testable by Chang’e 7 in August 2026. If an interface with $|R| > 0.02$ is detected within this window, the concentric stratification receives its first strong observational support, ahead of any SPH simulation. If no interface is detected under the falsification conditions stated, prediction P1 is

seriously called into question and the theory will require revision — something the author explicitly accepts.

This work invites the planetary science community to evaluate the TPT on its own terms: the equations are numbered, the derivations are explicit, the hypotheses are ranked, the limitations are acknowledged, and the predictions are falsifiable. The TPT will be judged primarily on its observational predictions — the seismic data from Chang’e 7 (August 2026) and the Artemis III samples (2028–2029) — ahead of and independently of any full numerical validation. The theory stands or falls on these results, not on the elaboration presented in this manuscript.

A Adiabatic temperature profile

With $T_{\text{surf}} \approx 3\,500$ K and an adiabatic gradient $dT/dr \approx -0.3$ K km⁻¹, the central temperature reaches $\approx 5\,000$ K. The peridotite solidus at 135 GPa is about 4 300 K (Stixrude & Lithgow-Bertelloni, 2014): melting is thermodynamically stable throughout the mantle, confirming Hypothesis H1.

B Demonstration of the $b' = 1$ attractor

The parameter b' is defined as the ratio between the critical ejection velocity U_{crit} and the geostrophic equilibrium velocity U_{eq} of the CMT:

$$b' = \frac{U_{\text{crit}}}{U_{\text{eq}}} = \frac{2\Omega \sin \epsilon \sqrt{2|g_{\text{eff}}| R_{\text{eq,proto}}}}{|g_{\text{eff}}|}. \quad (65)$$

Evolution due to Fe-Ni segregation. The density of the CMT decreases as $\rho_{\text{CMT}}(t) = \rho_0(1 - \zeta_{\text{Fe}}(t))$, where $\zeta_{\text{Fe}}(t)$ is the fraction of Fe-Ni segregated. The effective gravity $|g_{\text{eff}}|$ is proportional to ρ_{CMT} , so

$$\left. \frac{db'}{dt} \right|_{\text{segregation}} = \frac{1}{2} \frac{\dot{\rho}_{\text{CMT}}}{\rho_{\text{CMT}}} b' > 0, \quad (66)$$

since $\dot{\rho}_{\text{CMT}} < 0$: Fe-Ni segregation increases b' .

Evolution due to ejection. An ejection carries away a fraction $\epsilon = M_{\text{ej}}^{\text{max}}/M_{\text{CMT}} \approx 0.1$ of the CMT mass. The angular momentum decreases as $\Delta\Omega/\Omega \approx -\epsilon$, and the effective gravity decreases as $\Delta|g_{\text{eff}}|/|g_{\text{eff}}| \approx -\epsilon$ (proportional to the lost mass). Thus,

$$\left. \frac{\Delta b'}{b'} \right|_{\text{ejection}} = \frac{\Delta\Omega}{\Omega} - \frac{1}{2} \frac{\Delta|g_{\text{eff}}|}{|g_{\text{eff}}|} \approx -\epsilon + \frac{1}{2}\epsilon = -\frac{1}{2}\epsilon < 0 : \quad (67)$$

ejection decreases b' .

Stable fixed point. The two tendencies oppose each other: segregation increases b' ,

ejection decreases it. The equilibrium point $b' = 1$ is a stable attractor. For $b' < 1$, the system is unstable and an ejection occurs, further decreasing b' (since $\Delta b' < 0$); for $b' > 1$, the system is stable, no ejection occurs, and segregation pushes b' upward, toward 1, from below. The point $b' = 1$ is therefore a stable attractor of the coupled segregation–ejection system.

C Complete derivation of the coefficient λ

This appendix gives the complete analytical derivation of the stability coefficient $\lambda(U)$, with dimensional verification and, for each term, its physical story. It follows the Solberg-Høiland parcel-displacement formalism (Solberg, 1936; Høiland, 1941; Tassoul, 1978) adapted to the TPT geometry, and arrives at the form used in Section 9.2.

C.1 The stability equation

The idea behind the method can be summarized in one sentence: we mentally displace a small fluid parcel along the radius, tally the forces that pull it back or carry it away, and the sign of the balance tells us whether the torus is confined or ejected. For an infinitesimal radial displacement ξ_r of a CMT parcel:

$$\ddot{\xi}_r = \lambda(U) \xi_r, \quad \lambda > 0 \text{ (instability)}, \quad \lambda < 0 \text{ (stable confinement)}. \quad (68)$$

The displacement is *spherically* radial, and latitude is measured from the body’s instantaneous dynamical equator — no orbital (ecliptic) reference enters the internal dynamics, which only knows Ω and the local vertical. The angle ε_{dyn} denotes the *colatitude of the CMT* relative to this dynamical equator: $\sin \varepsilon_{\text{dyn}} = \cos \phi$, where ϕ is the torus’s latitude ($\varepsilon_{\text{dyn}} = 70^\circ \leftrightarrow \phi = 20^\circ$, at the heart of the intertropical band).

C.2 The Rayleigh (epicyclic) term

The displaced parcel conserves its specific angular momentum $L = rU$, while the background flow follows the profile $U(r) \propto r^\alpha$. Finding itself at a radius where the background rotates differently, the parcel experiences a restoring force: this is the Rayleigh mechanism (Rayleigh, 1917), whose strength is the squared epicyclic frequency,

$$\kappa^2 = \frac{1}{r^3} \frac{d(rU)^2}{dr} = \frac{2(1+\alpha) U^2}{r^2}, \quad (69)$$

obtained by differentiating $(rU)^2 \propto r^{2(1+\alpha)}$. This term is always positive (stabilizing) for any physical profile ($\alpha > -1$). Crucial point: this centrifugal-load gradient contains *by itself* the entire centrifugal effect. Adding a separate “centrifugal pressure” term would amount to double counting the same effect; the present derivation therefore contains none.

C.3 The non-traditional Coriolis component (Eötvös effect)

In a rotating frame, any prograde flow is radially *lightened*: the Coriolis force applied to the azimuthal velocity U has a component along the local vertical, $+2\Omega U \cos \phi$, directed outward. This is the Eötvös effect — the same one that makes an object lighter when moving eastward. The “traditional approximation” of geophysics neglects this component; it must be retained at low latitudes and for deep flows (Gerkema et al., 2008) — precisely the regime of the intertropical band. Writing $\sin \varepsilon_{\text{dyn}} = \cos \phi$:

$$a_{\text{Eötvös}} = + \frac{2\Omega \sin \varepsilon_{\text{dyn}}}{r} U \xi_r, \quad (70)$$

a destabilizing term: it opposes gravitational confinement.

C.4 The role of the Euler acceleration (nutation)

The rotation axis nutates, and one might be tempted to add to λ an Euler term from $-\dot{\Omega} \times \mathbf{r}$. This temptation must be resisted, for a structural reason: λ gathers accelerations *proportional to the displacement* ξ_r , whereas the Euler acceleration is a time-dependent *forcing*, independent of ξ_r (and a combination such as $|\dot{\Omega}| r / \Omega^2$ is moreover dimensionally in meters, not s^{-2}). Its physical place is elsewhere, and it is already occupied in the TPT: nutation slowly modulates the effective colatitude $\varepsilon_{\text{dyn}}(t)$ — the modulation parameter $|\dot{\Omega}| / \Omega^2 \approx 0.08 \sin \theta_{\text{nut}} \ll 1$ guarantees the adiabatic regime — and periodically lowers Ω_{crit} via the Eötvös term $2 \sin \varepsilon_{\text{dyn}}$. This is precisely the *parametric amplification* channel of Section 9.2: the Euler effect is real, maximal at nutation peaks, and enters the mechanism as a modulation of the threshold, not as a term in λ .

C.5 Buoyancy: thermal–compositional partition

The body’s density gradient has two sources, and the derivation separates them in the manner of the Ledoux criterion. The *thermal* part — the magma is cooled from above, hence superadiabatic — enters through the Brunt–Väisälä frequency (Väisälä, 1925; Brunt, 1927) along gravity:

$$-N^2 > 0 \quad (N^2 < 0, \text{ convection}), \quad (71)$$

a destabilizing contribution that sustains convection. The *compositional* part — Fe-Ni segregation makes the body denser at depth — is measured by $\beta \equiv -d \ln \rho / d \ln r > 0$ and enters through the *centrifugal* channel. The picture is simple: the centrifugal force is a gravity that points outward; a stratification stable for true gravity (dense at the bottom) is, for this inverted gravity, a Rayleigh-Taylor-type configuration, which pushes the dense fluid outward:

$$a_\beta = + \frac{\beta U^2}{r^2} \xi_r. \quad (72)$$

A destabilizing term — and, as we shall see, indispensable: it is the fuel of the engine. Fe-Ni segregation thus does double duty in the TPT: it accelerates rotation through contraction (skater effect) and it creates the stratification whose centrifugal buoyancy fuels the ejection.

C.6 Assembled coefficient and dimensional check

Summing the confining gravity $-g_{\text{eff}}/r$, the thermal buoyancy $-N^2$, the Eötvös effect (70), the Rayleigh restoring term (69), and the centrifugal buoyancy (72):

$$\lambda(U) = -\frac{g_{\text{eff}}}{r} - N^2 + \frac{2\Omega \sin \varepsilon_{\text{dyn}}}{r} U - \frac{2(1 + \alpha) - \beta}{r^2} U^2. \quad (73)$$

Each term is in s^{-2} : $[g_{\text{eff}}/r] = (\text{m s}^{-2})/\text{m} = \text{s}^{-2}$; $[N^2] = \text{s}^{-2}$; $[\Omega \sin \varepsilon_{\text{dyn}} U/r] = \text{s}^{-1}(\text{m s}^{-1})/\text{m} = \text{s}^{-2}$; $[U^2/r^2] = (\text{m s}^{-1})^2/\text{m}^2 = \text{s}^{-2}$.

C.7 Why is the Eötvös term destabilizing here?

A legitimate objection must be addressed directly. In the canonical derivation in cylindrical coordinates — displacement in the plane of rotation, conserved absolute angular momentum $j = r(U + \Omega r \sin \varepsilon_{\text{dyn}})$ — the Coriolis contribution is absorbed into the epicyclic frequency and becomes *stabilizing*. This absorption is exact at the strict equator. But the CMT is not an equatorial line: it is a band whose center of mass lies off the equator, and three established results change the picture there. (1) Off the equator, epicyclic absorption only acts on the cylindrical projection of the displacement, and the non-traditional component must be retained (Gerkema et al., 2008). (2) More importantly, the TPT's rotation varies on spheres ($U(r)$ in spherical radius) rather than on cylinders: it violates the second Goldreich–Schubert stability condition ($\partial\Omega/\partial z = 0$) (Goldreich & Schubert, 1967), so that there exist *tilted* displacement paths along which the Solberg–Høiland criteria are violated — non-diffusive instability as soon as $N^2 \leq 0$, which is the case here, *regardless* of the sign assigned to the radial Coriolis term. (3) The classical inviscid criteria further *underestimate* instability as soon as the Prandtl number departs from unity (McIntyre, 1970), which is the case for silicate magma. Finally, the latitudinal shear at the edge of the torus, $\partial U/\partial y \sim U/W$, is comparable to 2Ω (ratio ≈ 0.95 at nominal values): the CMT operates near the equatorial inertial instability criterion (Dunkerton, 1981). The geometric condition of the present criterion — unfavorable absolute angular momentum gradient along tilted paths at the edge of the CMT — is verifiable by 3D SPH simulation and connects to Limitation L14.

C.8 The rotation threshold

Setting $\lambda = 0$ and substituting $U = \Omega r$:

$$\Omega_{\text{crit}}^2 = \frac{g_{\text{eff}}/r + N^2}{2 \sin \varepsilon_{\text{dyn}} - 2(1 + \alpha) + \beta}. \quad (74)$$

A finite threshold exists only if the denominator D is positive, i.e., $\beta > 2(1 + \alpha) - 2 \sin \varepsilon_{\text{dyn}}$. The driving role of centrifugal buoyancy is visible here: at nominal values, $2 \sin \varepsilon_{\text{dyn}} - 2(1 + \alpha) = 1.879 - 3 = -1.12 < 0$ — without β , no rotation would ever trigger ejection. With $\varepsilon_{\text{dyn}} = 70^\circ$ at the nutation peak, $\alpha = 0.5$, $\beta = 2.5$, and $N^2 = -1.93 \times 10^{-7} \text{ s}^{-2}$, the denominator is $D = 1.379$ and the threshold $\Omega_{\text{crit}} \approx 1.27 \Omega_0$. Note that the present form, freed of double counting, halves the threshold's sensitivity to the velocity profile ($\partial D / \partial \alpha = -2$ instead of -4), and coincides numerically with the earlier form at the point $\alpha = 0.5$. This threshold is a well-defined surface in the parameter space $(\Omega, \varepsilon_{\text{dyn}}, \beta)$, giving the mechanism a predictive character: ejection occurs when $\Omega > \Omega_{\text{crit}}$, the crossing operating through internal, angular-momentum-conserving channels — segregation contraction (skater effect) and parametric amplification (Section 9.2) — the Late Veneer's impulsive trigger acting only as a modulation at the nutation peak.

D Demonstration of the Bi_{KH} criterion

The Bingham-Kelvin-Helmholtz number compares the yield stress to the KH shear stress:

$$\text{Bi}_{\text{KH}} = \frac{\tau_y k}{\rho \sigma_{\text{KH}} \Delta U}, \quad (75)$$

where k is a geometric factor (~ 1), $\sigma_{\text{KH}} \sim \Delta U / \lambda$ is the KH growth rate, and λ is the wavelength of the perturbation. Thus,

$$\text{Bi}_{\text{KH}} \sim \frac{\tau_y \lambda}{\rho \Delta U^2}. \quad (76)$$

For the CMT: $\tau_y \sim 10^2\text{--}10^4 \text{ Pa}$, $\lambda \sim R_{\text{torus}} \sim 10^7 \text{ m}$, $\rho \sim 3000 \text{ kg m}^{-3}$, $\Delta U \sim U_{\text{crit}} \sim 10^4 \text{ m s}^{-1}$. This gives $\text{Bi}_{\text{KH}} \sim 3 \times 10^{-3}\text{--}3 \times 10^{-1}$.

Effective cohesion is the product of three independent reinforcements (atmospheric confinement, internal crystal network, Rayleigh-Taylor suppression), detailed in Section 9.1:

$$\text{Bi}_{\text{KH}}^{\text{eff}} = \text{Bi}_{\text{KH}} \times f_{\text{atm}} \times f_{\text{crist}} \times f_{\text{RT}}, \quad (77)$$

with $f_{\text{atm}} \sim 10\text{--}100$ (atmospheric confinement, $P \sim 10^5\text{--}10^7 \text{ Pa}$), $f_{\text{crist}} \sim 3\text{--}10$ (internal crystal network), and $f_{\text{RT}} \sim 10\text{--}50$ (Rayleigh-Taylor suppression, $\rho_{\text{flow}} / \rho_{\text{atm}} \sim 200$). With nominal values, $\text{Bi}_{\text{KH}} \sim 3 \times 10^{-2}$ and the product of the three reinforce-

ments is ~ 300 , giving:

$$\text{Bi}_{\text{KH}}^{\text{eff}} \sim 9, \quad (78)$$

much greater than 1. Cohesion is therefore assured.

Acknowledged limitation: this calculation is a nominal estimate. A 3D SPH simulation with full BH rheology is required to validate the exact values, in particular the nonlinear interaction among the three effects (Limitation L14).

E Derivation of the Elsasser number and of B_{crit}

Balancing the Coriolis force $F_C \sim 2\rho\Omega U$ and the Lorentz force $F_L \sim B^2/(\mu L)$, with $\eta \sim UL$ (unit-order magnetic Reynolds number at the dynamo threshold):

$$B^2 \sim 2\mu\rho\Omega\eta, \quad \Rightarrow \quad \Lambda \equiv \frac{B^2}{\rho\mu\eta\Omega} \sim 2, \quad (79)$$

i.e., $\Lambda \sim \mathcal{O}(1)$, consistent with the values $\Lambda \sim 0.1\text{--}1$ inferred for the present-day terrestrial core (Christensen & Aubert, 2006). Using $\eta = (\mu_0\sigma)^{-1}$ with $\sigma \approx 1.5 \times 10^6 \text{ S m}^{-1}$, a representative electrical conductivity of liquid iron under core conditions (Pozzo et al., 2012), $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$, $\rho \approx 10^4 \text{ kg m}^{-3}$, $\Omega \approx 4.99 \times 10^{-4} \text{ rad s}^{-1}$, we obtain $B_{\text{crit}} \approx 2.6 \text{ mT}$, of the same order of magnitude as the field inferred within the present-day terrestrial core (Christensen & Aubert, 2006). Note that this value assumes Ω at its formation value ($T = 3.5 \text{ h}$). At dynamo ignition ($t \approx 330\text{--}350 \text{ Ma}$), Ω has decreased through two cumulative effects: the loss associated with the three ejection episodes themselves ($\Delta\Omega/\Omega \approx -10$ to -15% per episode, Section 9.2), followed by additional drainage through the evection resonance (Section 11.5), by a factor of ≈ 2 . Cascading these two effects, $\Omega_{\text{dynamo}}/\Omega_0 \approx 0.28\text{--}0.35$; since $B_{\text{crit}} \propto \sqrt{\Omega}$, this gives $B_{\text{crit}} \approx 1.4\text{--}1.5 \text{ mT}$ — same order of magnitude, conclusion unchanged, and still consistent with geodynamo models.

F Stokes sedimentation timescale of Fe-Ni

Stokes sedimentation velocity for a Fe-Ni droplet of radius $r \approx 5 \text{ cm}$ in a molten silicate:

$$v_{\text{Stokes}} = \frac{2(\rho_{\text{Fe}} - \rho_{\text{sil}})g_{\text{eff}}^{(0)}r^2}{9\eta} \approx \frac{2 \cdot 3300 \times 6.42 \times (0.05)^2}{9 \cdot 0.1} \approx 118 \text{ m s}^{-1}. \quad (80)$$

Over a characteristic sedimentation distance $\ell \approx 30 \text{ km}$: $\tau_{\text{Stokes}} = \ell/v_{\text{Stokes}} \approx 254 \text{ s}$. Robust for $r \in [1, 10] \text{ cm}$ and $\eta \in [0.05, 1] \text{ Pa s}$.

G Accretion of magmatic ejecta: why perfect cohesion is not required

A recurring objection to the TPT mechanism concerns the cohesion of the ejected CMT flow during transit between the surface of the proto-Earth and the Roche radius. This appendix shows that, even without perfect flow cohesion, the physical properties of the ejected magma and orbital mechanics guarantee accretion into a single body.

Magma is not dust. The ejected matter is a silicate magma at $T \approx 3\,000\text{--}4\,000\text{ K}$. Its dynamic viscosity is $\eta \approx 1\text{--}100\text{ Pa s}$ (Giordano et al., 2008) — comparable to hot honey. Unlike solid rock fragments or a gas cloud, a high-temperature magma does not disperse: it deforms and stretches, but remains dense and cohesive at the macroscopic scale. Two sticky drops of magma at low relative velocity do not bounce off each other — they merge. Viscous coalescence replaces mechanical cohesion as the dominant aggregation mechanism once the flow has fragmented.

Orbital concentration. The ejecta of a given TPT episode share, to first order, the same initial orbital energy budget. By conservation of energy and angular momentum, they populate a band of neighboring orbits rather than dispersing isotropically over 4π steradians. This orbital concentration is fundamentally different from an isotropic explosion: it is the same physical principle that produces accretion disks from tidally disrupted bodies. The CMT's toroidal geometry reinforces this effect: ejection occurs preferentially within the oblique intertropical band $|\phi| < 30^\circ$, concentrating the ejecta within a limited azimuthal solid angle. The resulting debris field is geometrically narrow, maximizing mutual encounter rates.

Beyond the Roche radius: accretion is inevitable. Beyond the Roche radius $R_{\text{Roche}} = 2.456R_\oplus$, the self-gravitation of the ejecta cloud overcomes tidal disruption. For co-orbital magma fragments of total mass $M_{\text{cloud}} \approx 2\text{--}4 \times 10^{22}\text{ kg}$, the gravitational accretion timescale is:

$$\tau_{\text{acc}} \approx \frac{1}{\sqrt{G\rho_{\text{cloud}}}} \approx 10^3\text{--}10^5\text{ years}, \quad (81)$$

much shorter than the thermal lifetime of the magma ($\tau_{\text{cool}} \sim 10^6\text{--}10^7$ years for a Moon-sized body, Elkins-Tanton 2008). During this accretion window, the magma remains above its solidus: every collision is plastic, not elastic. There is neither a fragmentation cascade nor bounce loss. The capture efficiency f_{cap} therefore has a physical lower bound independent of the BH cohesion hypothesis:

$$f_{\text{cap}}^{\text{min}} \approx \frac{M_{\text{cloud}}(r > R_{\text{Roche}})}{M_{\text{ej}}^{\text{total}}} \gtrsim 0.45\text{--}0.55, \quad (82)$$

the lost fraction ($\approx 45\text{--}55\%$) corresponding to ejecta that fall back onto the proto-

Earth or escape on hyperbolic orbits.

Implication for the theory. The capture efficiency $f_{\text{cap}} = 0.70$ adopted in the main text is therefore not a free parameter that the theory must defend at all costs. Even in the pessimistic scenario in which the CMT flow fragments completely upon ejection, orbital mechanics and magma viscosity guarantee the accretion of the co-orbital debris cloud beyond the Roche radius. The critical question shifts from “does the flow remain cohesive?” to “what fraction of the ejecta reaches the Roche radius?” — a question that depends on the ejecta velocity distribution, not on BH rheology. This reformulation makes the theory more robust: cohesion is advantageous, but not necessary.

H Complementary mechanisms for the ejection threshold

This appendix identifies the physical mechanisms that contribute to reaching the ejection threshold $\lambda(U) = 0$. The threshold is a condition on Ω :

$$\Omega > \Omega_{\text{crit}} \approx \sqrt{\frac{|g_{\text{eff}}|}{2 \sin \varepsilon r}} \cdot \sqrt{\frac{1 + \alpha}{r}}. \quad (83)$$

With the nominal proto-Earth parameters ($\Omega_0 = 4.99 \times 10^{-4}$ rad/s, $\varepsilon = 55^\circ$), the threshold is $\Omega_{\text{crit}} \approx \sqrt{2} \Omega_0$.

The angular momentum input needed to reach this threshold is supplied by continuous planetesimal bombardment. Late accretion (the “Late Veneer”) is delivered by a small number of large bodies (radius $\gtrsim 750$ km) (Raymond et al., 2013). Geodynamic constraints on the delivery of highly siderophile elements to the mantle — impactors larger than ~ 1 km deliver metal that sinks toward the core rather than remaining entrained in the convecting mantle — push the total late-accretion mass up to $\sim 5\%$ of Earth’s mass (Anslow et al., 2026). Assigning to this mass a characteristic impact velocity $v_{\text{imp}} \sim 10 \text{ km s}^{-1}$ and a lever arm of order r , the cumulative angular momentum available is:

$$L_{\text{imp}} \sim M_{\text{imp}} v_{\text{imp}} r \approx 2.2 \times 10^{34} \text{ J s}. \quad (84)$$

This input is sufficient to raise Ω from Ω_0 to Ω_{crit} in the geometrically most favorable case ($\varepsilon \approx 90^\circ$), and provides the margin needed for ejection to occur at the peaks of the nutation cycle. The Euler oscillation term λ_4 modulates the effective obliquity ε , but does not provide the main driving force.

An external dynamical cause for bombardment: the early giant-planet instability

Appendix H assumes the availability of intense bombardment by large bodies during the active TPT window (30–50 Ma), without identifying its dynamical cause. Recent

work provides such a cause, independent of the proto-Earth's internal thermodynamics.

The classical narrative placed the orbital instability of the giant planets (analogous to the "Nice model") late, around 3.9 Ga, linked to the lunar Late Heavy Bombardment (LHB). More recent dynamical and geochronological work challenges this late chronology: the instability is instead argued to be triggered by the dispersal of the gaseous protoplanetary disk itself, and would therefore occur *early*, a few to ten million years after Solar System formation (Nesvorný et al., 2022; Mojzsis et al., 2019). Mojzsis et al. (2019) show that this early migration produces an intense influx of comets and small bodies over ~ 30 Myr, bounded to $\lesssim 4480$ Ma by the absence of post-dating reset ages for the most retentive chronometers of Earth, the Moon, Mars, and several meteorite parent bodies.

This ~ 30 Myr window of intense impactor influx, motivated by a purely dynamical cause (giant-planet orbital instability), substantially overlaps the active TPT window (30–50 Ma, Section 4.4), itself constrained by independent thermal considerations (silicate-atmosphere opacity, Hf-W chronometer). This convergence between two time windows derived from entirely distinct physics — gaseous disk dispersal on one hand, mantle melting thermodynamics on the other — constitutes an additional plausibility argument for the timely availability of the bombardment that Appendix H requires to cross the ejection threshold Ω_{crit} .

Acknowledged limitation: as with Section 11.5.2, this temporal convergence remains a comparison of compatible timescales, not a coupled dynamical simulation. It does not demonstrate that impactors from this instability actually reached the proto-Earth with the sizes ($\gtrsim 750$ km) and timing required by the ejection cycle; it provides a plausible dynamical mechanism where the main text simply assumed the availability of bombardment.

I Persistence of the post-ejection oscillation and repeated opportunities

The ejection threshold is not a single-pass condition. After an ejection, Ω falls below Ω_{crit} , but the system does not immediately come to rest. The cavity left by the ejection is bounded by the confining atmosphere and by the residual 'aā crust of the CMT. These boundaries are reflective rather than absorbing (as established by the $\text{Bi}_{\text{KH}} \gg 1$ criterion, Appendix D), so that post-ejection nutation persists over several cycles.

The obliquity excursion envelope $\varepsilon_{\text{exc}}(t)$ follows a two-timescale regime:

$$\varepsilon_{\text{exc}}(t) \approx \begin{cases} \varepsilon_{\text{exc}}^{(0)} \cos(\omega_{\text{nut}}^{(\text{post})} t) & 0 \leq t \lesssim t_{\text{rec}} \\ \varepsilon_{\text{exc}}^{(0)} e^{-(t-t_{\text{rec}})/\tau_{\text{BH}}} \cos(\omega_{\text{nut}}^{(\text{post})} t) & t \gtrsim t_{\text{rec}} \end{cases} \quad (85)$$

where $\omega_{\text{nut}}^{(\text{post})} = 2\pi/T_{\text{nut}}^{(\text{post})}$, with $T_{\text{nut}}^{(\text{post})} \approx 30$ h, and $\tau_{\text{BH}} \sim 10^2\text{--}10^4$ s (Eq. 1) for the decay that takes over once the plateau ends.

Over the duration of the recharge window, $t_{\text{rec}} \approx 10\text{--}12$ days, this plateau covers:

$$N_{\text{opp}}^{(\Delta)} = \frac{t_{\text{rec}}}{T_{\text{nut}}^{(\text{post})}} \approx \frac{240\text{--}288 \text{ h}}{30 \text{ h}} \approx 8\text{--}10 \quad (86)$$

complete cycles, each transiently bringing ε back near its post-shock maximum excursion. The ejection threshold is therefore not a single attempt but a modest number of repeated opportunities, each with a high individual probability.

Implication for the ejection mechanism. The post-ejection plateau window is a period of $\approx 8\text{--}10$ nutation cycles (about 10–12 days) during which ε remains close to its post-shock maximum. A major Late Veneer impact occurring during this window benefits from a double advantage:

1. ε is already close to its maximum, so Ω_{crit} is minimal;
2. The impact's impulsive torque provides a jump $\Delta\Omega$ added to Ω , already close to the threshold.

The combination of the two erases the threshold and triggers the next ejection. This mechanism links ejection episodes to late impacts (Late Veneer), rather than to simple continuous angular momentum accumulation. The isotopic signature of these late impacts (siderophile excess) should be present in the outermost lunar layers (Prediction P22).

J List of abbreviations

BH	Bingham-Herschel, rheological law.
BiKH	Bingham-Kelvin-Helmholtz number.
CME	Coronal Mass Ejection.
CMT	Coherent Magmatic Torus.
ε_{dyn}	Colatitude of the CMT relative to the instantaneous dynamical equator ($\sin \varepsilon_{\text{dyn}} = \cos \phi$).
ε_{orb}	Spin-orbit angle (orbital obliquity).
FSS	Farside Seismic Suite.
Hf-W	Hafnium-tungsten chronometer.
LHB	Late Heavy Bombardment (≈ 3.9 Ga).
LEMS	Lunar Environment Monitoring Station.
LMO	Lunar Magma Ocean.

NLS	Nonlinear Schrödinger equation.
POB	Proto-Orbital Body.
SPA	South Pole–Aitken (basin).
SPH	Smoothed Particle Hydrodynamics.
TPT	Triple Phase Transition.

Declarations

Availability of data and materials. All data generated or analyzed during this study are included in this published article (and its appendices). The interactive simulation accompanying this work is freely accessible at https://orion4622.github.io/moon-formation-triple-phase-transition/Animation_Formation_de_la_Lune_v2.html.

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